

Communicating with Data – Advanced Tableau

Dr. Ab Mosca (they/them)

Slides based off slides courtesy of Jordan Crouser (<https://jcrouser.github.io/>)

Notes

- Great job getting the physical visualization set up! There's already been a lot of interaction with it!
- Data Collection/Cleaning and Clean-up Teams, please check Slack for job details (copied below)

We need to clean up our display from the library by Nov. 13.

Data Collection Leads: Please plan to stop by the library and transcribe the data from your class's visualization into a data table **before Nov. 12**. You can pick the specific date/time amongst yourselves. Do your best to record what you see even if it is missing/nonsensical/doesn't follow the instructions; the data cleaning team will make final decisions on what to keep and correct.

Data Cleaning Leads: Please get the data table from the Data Collection Leads and clean it. Decide what to do with entries that do not make sense, and how to treat missing data. You can pick the specific date/time amongst yourselves, but please have data Nov. 14th.

Clean up Team: Please plan to stop by the library **on Nov. 12 or 13th** to clean up your class's display. You can pick the specific date/time amongst yourselves. Class is canceled on the 13th so you might consider using class time for clean-up. You can bring the visualizations and all materials to my office (I will leave the door unlocked for you). Please be thorough in your clean up so that we can continue to use the library for this project!

You are welcome to send any questions you have here and are encouraged to communicate via group chats or creating a new slack channel.

Reminder

- Final Project Instructions on the course website
- Find a group and start working on your project proposal

Plan for Today

- Working with multiple datasets
- Customizing the look of a visualization
- Working with filters
- Conditional backgrounds
- Details on demand (external resources)

Want to follow
along?

- Everything used in this demo is available in superhero-demo.zip under in-class on the course website

Running
example:
superheroes



First dataset: superdata.csv

Connections

superdata
Text file

Files

Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

superdata.csv

superhero-wikis.csv

New Union

New Table Extension

superdata

superdata.csv

superdata.csv 9 fields 17 rows

Name

superdata.csv

Fields

Type	Field Name	Physical Table	Remote Field Name
Abc	Name	superdata.csv	Name
Abc	Alignment	superdata.csv	Alignment
#	Intelligence	superdata.csv	Intelligence
#	Strength	superdata.csv	Strength
#	Speed	superdata.csv	Speed
#	Durability	superdata.csv	Durability
#	Power	superdata.csv	Power
#	Combat	superdata.csv	Combat
Abc	Universe	superdata.csv	Universe

Connection

Live Extract

Filters 0 Add

Abc superdata.csv Name	Abc superdata.csv Alignment	# superdata.csv Intelligence	# superdata.csv Strength	# superdata.csv Speed	# superdata.csv Durability
Aquaman	good	63	80	50	80
Batman	good	100	18	27	42
Booster Gold	good	56	85	53	50
Captain America	good	63	19	35	56
Captain Marvel	good	100	100	67	95
Deadpool	neutral	50	15	30	100
Flash	good	63	10	100	50
Iron Fist	good	75	23	35	42
Iron Man	good	100	85	58	85
Loki	bad	88	57	47	85
Red Tornado	good	75	38	67	60
Robin	good	88	11	27	28
Spider-Man	good	88	55	60	74

Data Source

Sheet 1

Navigation icons

Adding more datasets

Tableau - Book1

Connections [Add](#)

superdata
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

☒ superdata.csv

☒ superhero-wikis.csv

☐ New Union

☐ New Table Extension

Add a Connection
Add a new connection to use cross-database joins

Search for Data

Tableau Server

To a File

- Microsoft Excel
- Text file
- JSON file
- PDF file
- Spatial file
- Statistical file
- More...

To a Server

- Microsoft SQL Server
- MySQL
- Oracle
- Amazon Redshift
- More...

Connection ☒ Live ☐ Extract

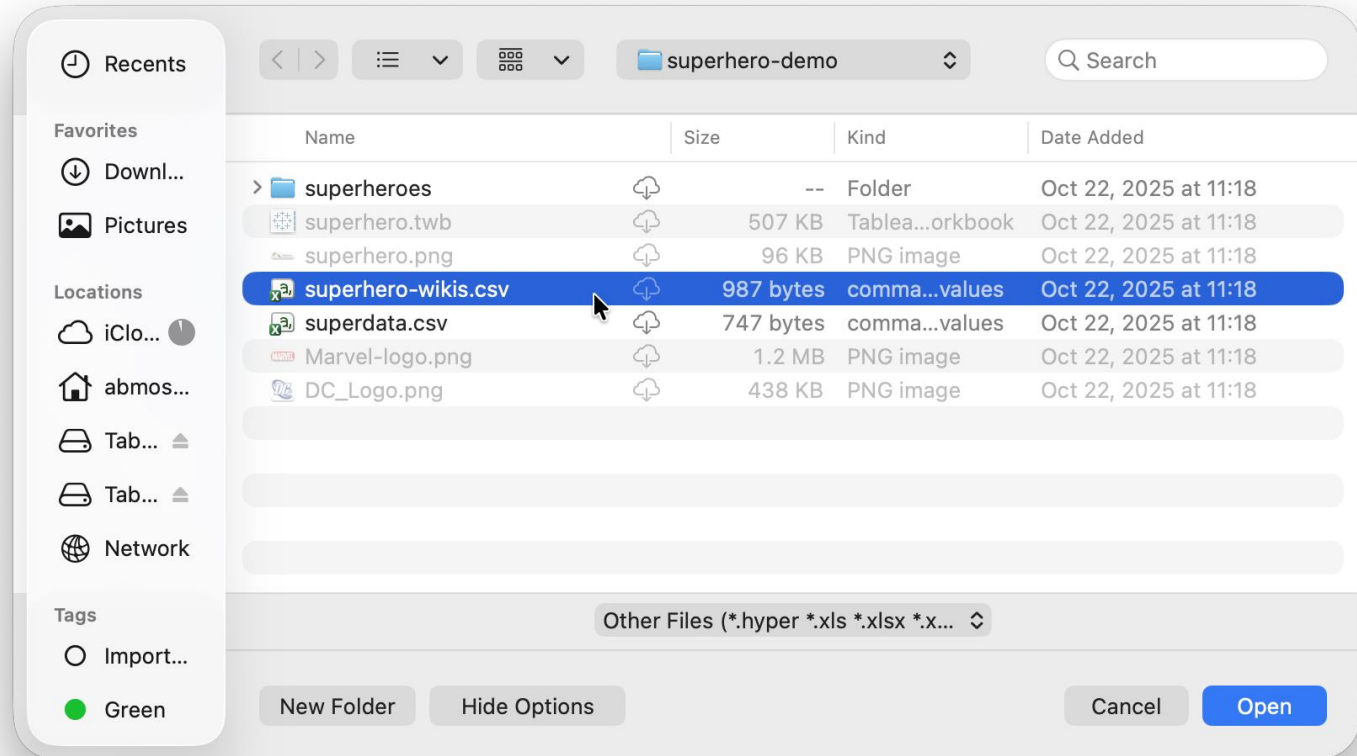
Filters 0 | [Add](#)

17 rows

Abc superdata.csv	Abc superdata.csv	# superdata.csv	# superdata.csv	# superdata.csv	# superdata.csv
Name	Alignment	Intelligence	Strength	Speed	Durability
Aquaman	good	63	80	50	
Batman	good	100	18	27	
Booster Gold	good	56	85	53	
Captain America	good	63	19	35	
Captain Marvel	good	100	100	67	
Deadpool	neutral	50	15	30	
Flash	good	63	10	100	
Iron Fist	good	75	23	35	
Iron Man	good	100	85	58	
Loki	bad	88	57	47	
Red Tornado	good	75	38	67	
Robin	good	88	11	27	
Spider-Man	good	88	55	60	

Data Source Sheet 1

Adding more datasets



Second dataset: superhero-wikis.csv

Tableau - Book1

superdata.csv (Multiple Connections)

Connection
☒ Live ☐ Extract

Filters
0 | Add

Connections

superdata
Text file

superhero-wikis
Text file

Files

☐ Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

superdata.csv

superhero-wikis.csv

New Union

New Table Extension

superdata.csv

9 fields 17 rows

17 rows

Name
superdata.csv

Type	Field Name	Physical Table	Remote Field Name
Abc	Name	superdata.csv	Name
Abc	Alignment	superdata.csv	Alignment
#	Intelligence	superdata.csv	Intelligence
#	Strength	superdata.csv	Strength
#	Speed	superdata.csv	Speed
#	Durability	superdata.csv	Durability
#	Power	superdata.csv	Power
#	Combat	superdata.csv	Combat
Abc	Universe	superdata.csv	Universe

Abc superdata.csv	Abc superdata.csv	# superdata.csv	# superdata.csv	# superdata.csv	# superdata.csv
Name	Alignment	Intelligence	Strength	Speed	Durability
Aquaman	good	63	80	50	
Batman	good	100	18	27	
Booster Gold	good	56	85	53	
Captain America	good	63	19	35	
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Iron Fist	good	75	23	35	
Iron Man	good	100	85	58	
Loki	bad	88	57	47	
Red Tornado	good	75	38	67	
Robin	good	88	11	27	
Spider-Man	good	88	55	60	

Relationships vs. Joins

- Default method in Tableau Desktop is to use relationships to combine data sources
- Relationships:
 - preserve original table's level of detail
 - Do not require you to specify matching fields
 - Are automatic and context-aware
 - Flexible and custom based on what you are visualizing

Relating Tables

Connections

superdata

Text file

superhero-wikis

Text file

Files

☐ Use Data Interpreter

Data Interpreter might be able to clean your Text file workbook.

superdata.csv

superhero-wikis.csv

New Union

New Table Extension

superdata.csv + (Multiple Connections)

Connection

Live

Extract

Filters

0

Add

Relationship

Related Fields

superdata.csv * - * superhero-wikis.csv Name = Name (superhero-wiki...

superdata.csv

superhero-wikis.csv

superdata.... — superhero...

17 rows

How do relationships differ from joins? [Learn more](#)

superdata.csv

Operator

superhero-wikis.csv

Abc Name = Abc Name (superher...

+

 Add more fields

> Performance Options

Remove Relationship

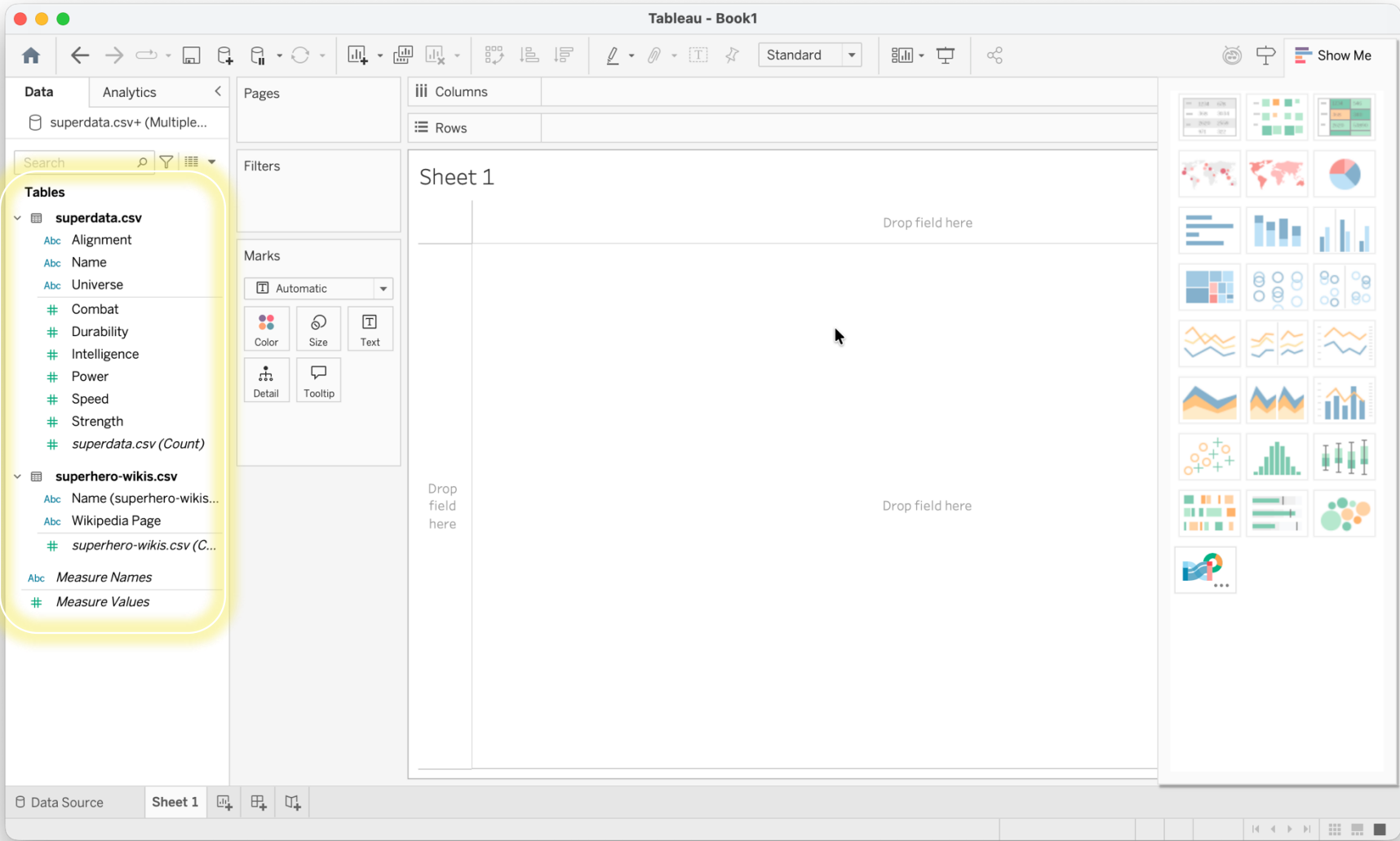
Abc superhero-wikis.csv Name (superhero-wikis.cs	Abc superhero-wikis.csv Wikipedia Page
Aquaman	https://en.m.wikipedia.org/wi...
Batman	https://en.m.wikipedia.org/wi...
Booster Gold	https://en.m.wikipedia.org/wi...
Captain America	https://en.m.wikipedia.org/wi...
Captain Marvel	https://en.m.wikipedia.org/wi...
Deadpool	https://en.m.wikipedia.org/wi...
Flash	https://en.m.wikipedia.org/wi...
Iron Fist	https://en.m.wikipedia.org/wi...

Joining the two

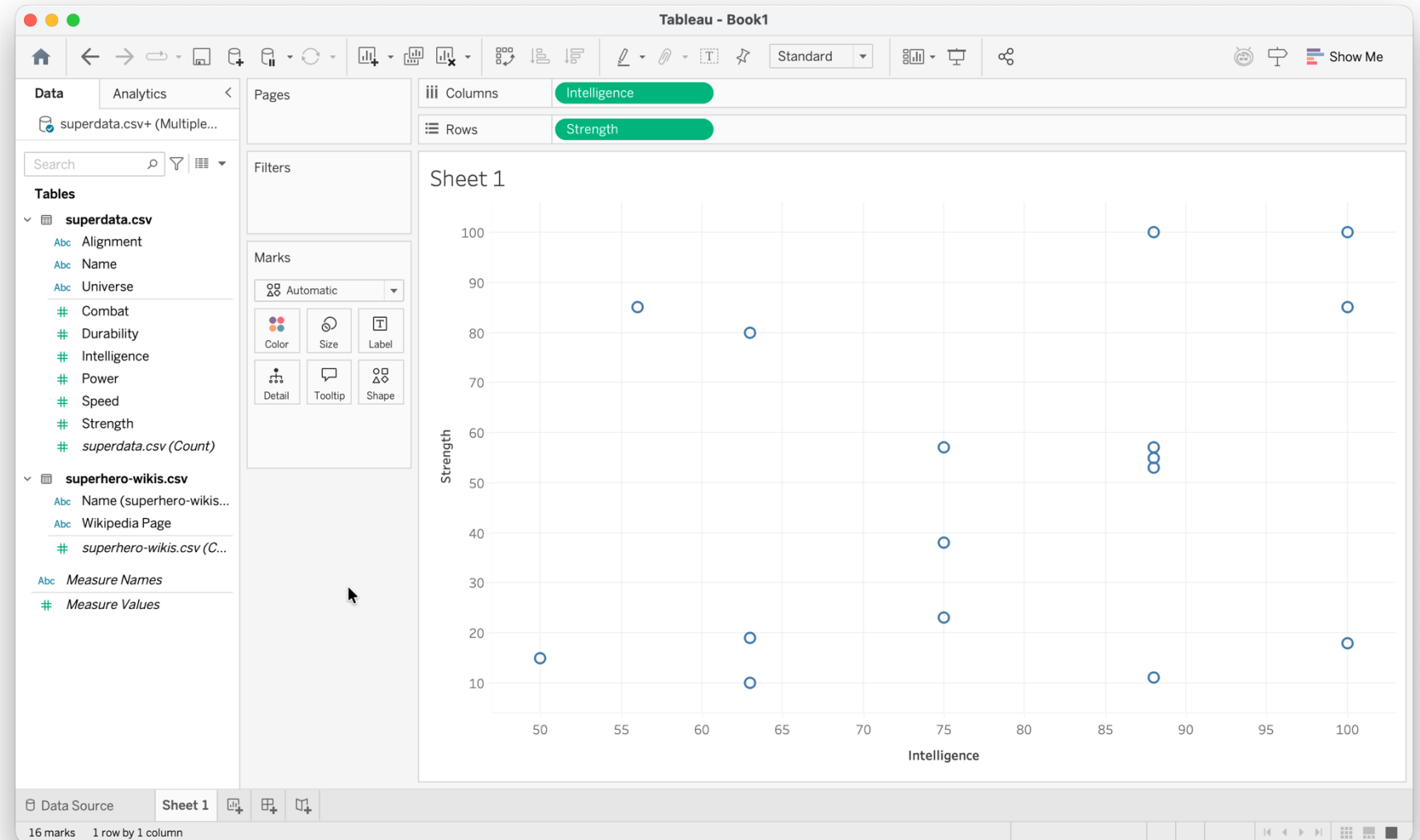
	A	B	C	D	E	F	G	H	I
1	Name	Alignment	Intelligence	Strength	Speed	Durability	Power	Combat	Universe
2	Aquaman	good	63	80	50	80	65	80	DC
3	Batman	good	100	18	27	42	37	100	DC
4	Booster Gold	good	56	85	53	50	72	40	DC
5	Captain Ame	good	63	19	35	56	46	100	Marvel
6	Captain Marv	good	100	100	67	95	62	56	DC
7	Deadpool	neutral	50	15	30	100	100	100	Marvel
8	Flash	good	63	10	100	50	57	32	DC
9	Iron Fist	good	75	23	35	42	43	100	Marvel
0	Iron Man	good	100	85	58	85	100	64	Marvel
1	Loki	bad	88	57	47	85	85	56	Marvel
2	Red Tornado	good	75	38	67	60	92	40	DC
3	Robin	good	88	11					
4	Spider-Man	good	88	55					
5	Superman	good	100	100					
6	Venom	bad	75	57					
7	Wonder Wor	good	88	100					
8	X-Man	good	88	53					

	A	B	C	D	E	F
1	Name	Wikipedia Page				
2	Aquaman	https://en.m.wikipedia.org/wiki/Aquaman				
3	Batman	https://en.m.wikipedia.org/wiki/Batman				
4	Booster Gold	https://en.m.wikipedia.org/wiki/Booster_Gold				
5	Captain Ame	https://en.m.wikipedia.org/wiki/Captain_America				
6	Captain Marv	https://en.m.wikipedia.org/wiki/Captain_Marvel_%28DC_Comics%29				
7	Deadpool	https://en.m.wikipedia.org/wiki/Deadpool				
8	Flash	https://en.m.wikipedia.org/wiki/Flash_%28comics%29				
9	Iron Fist	https://en.m.wikipedia.org/wiki/Iron_Fist_%28comics%29				
10	Iron Man	https://en.m.wikipedia.org/wiki/Iron_Man				
11	Loki	https://en.m.wikipedia.org/wiki/Loki_%28comics%29				
12	Red Tornado	https://en.m.wikipedia.org/wiki/Red_Tornado				
13	Robin	https://en.m.wikipedia.org/wiki/Robin_%28comics%29				
14	Spider-Man	https://en.m.wikipedia.org/wiki/Spider-Man				
15	Superman	https://en.m.wikipedia.org/wiki/Superman				
16	Venom	https://en.m.wikipedia.org/wiki/Venom_%28comics%29				
17	Wonder Wor	https://en.m.wikipedia.org/wiki/Wonder_Woman				
18	X-Man	https://en.m.wikipedia.org/wiki/Nate_Grey				

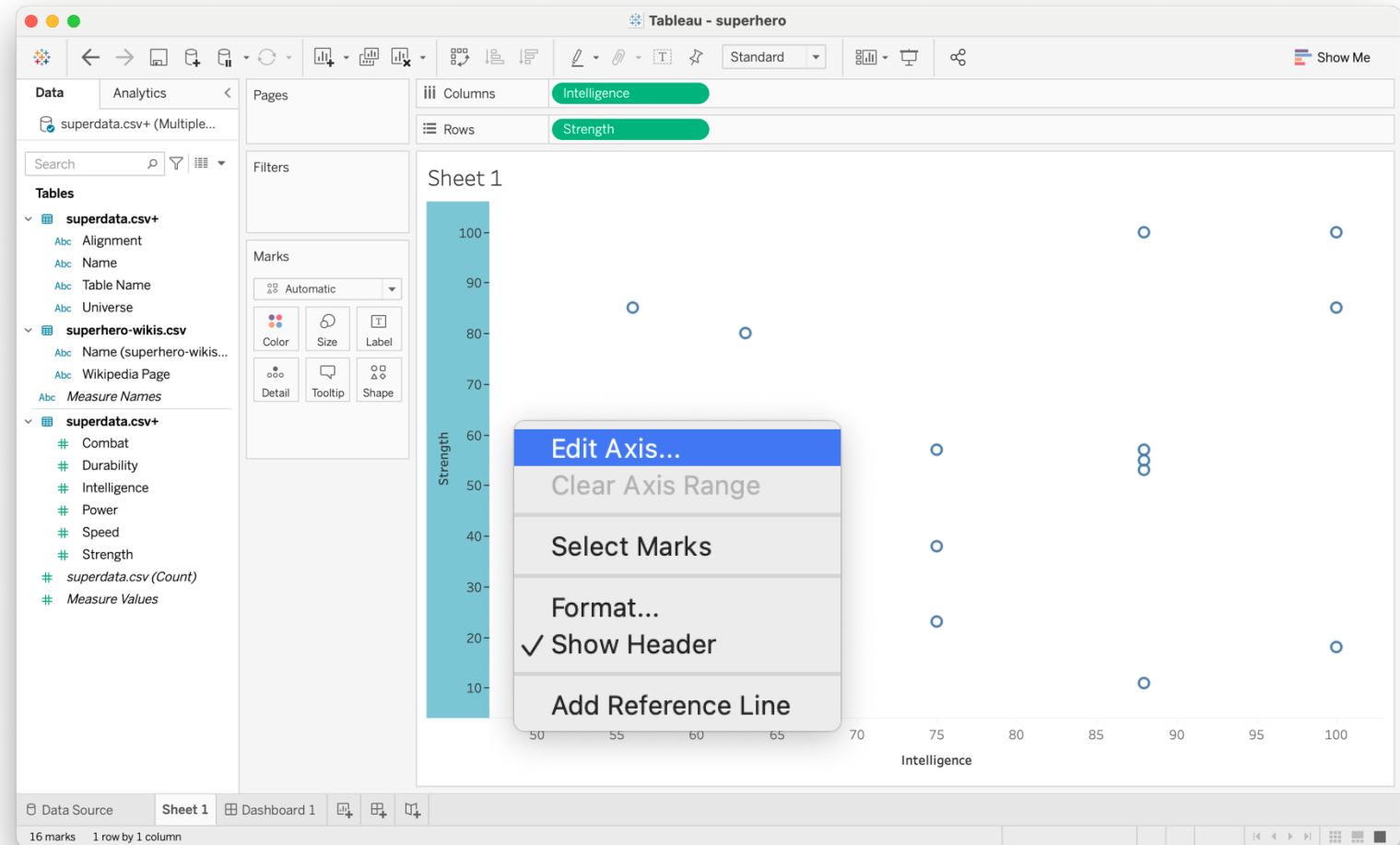
Over to Sheet1



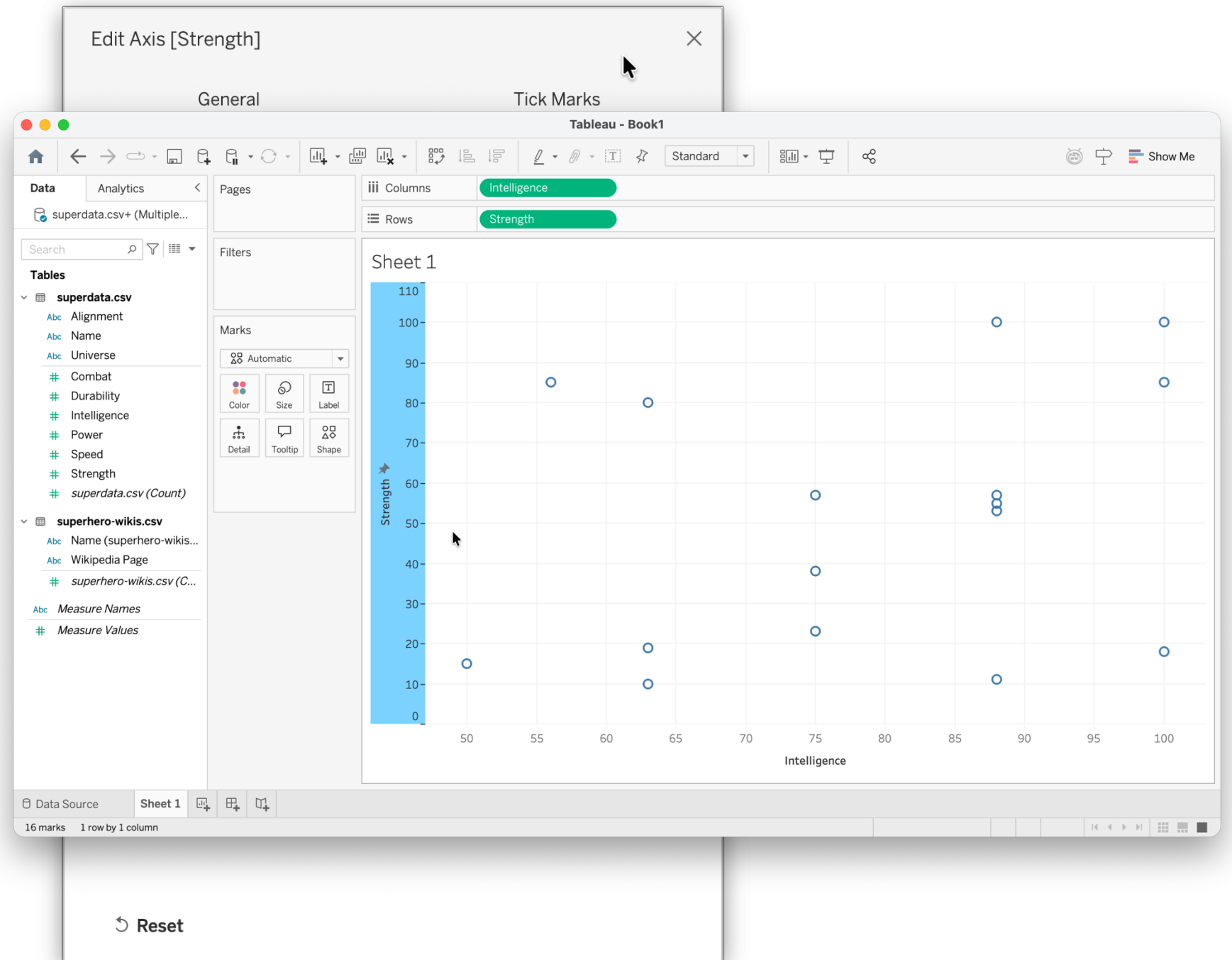
Simple scatterplot



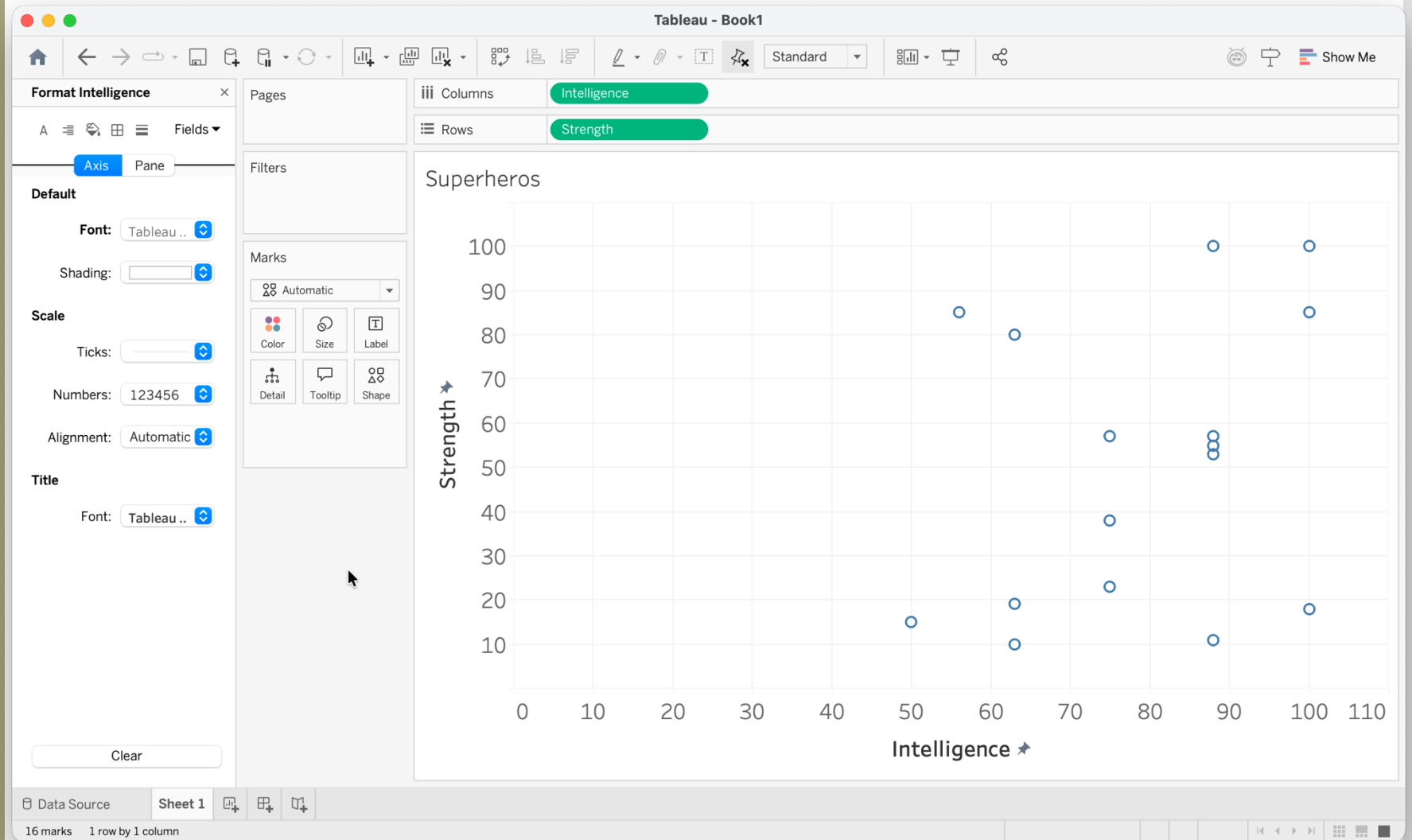
Editing axis values



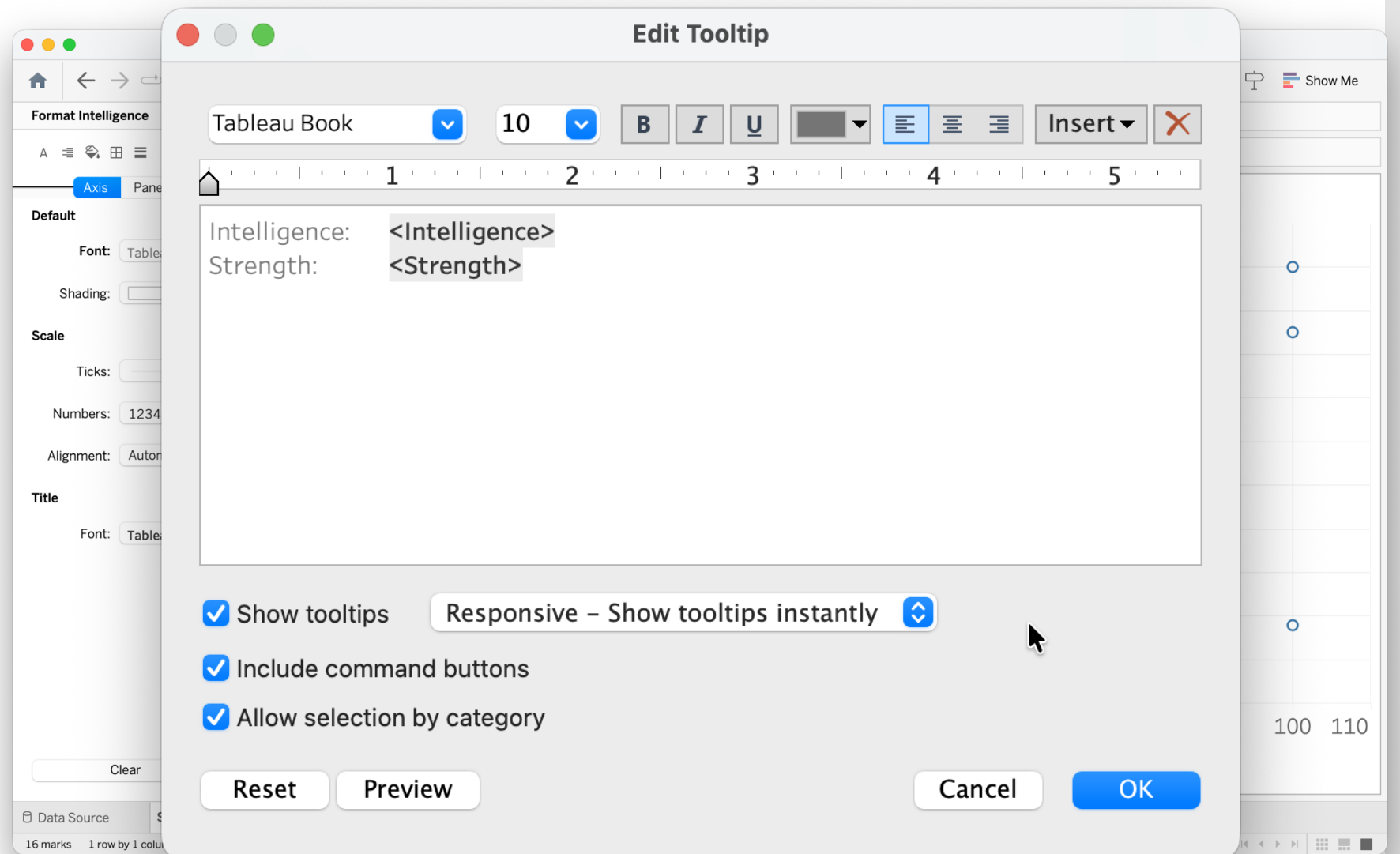
Editing axis values



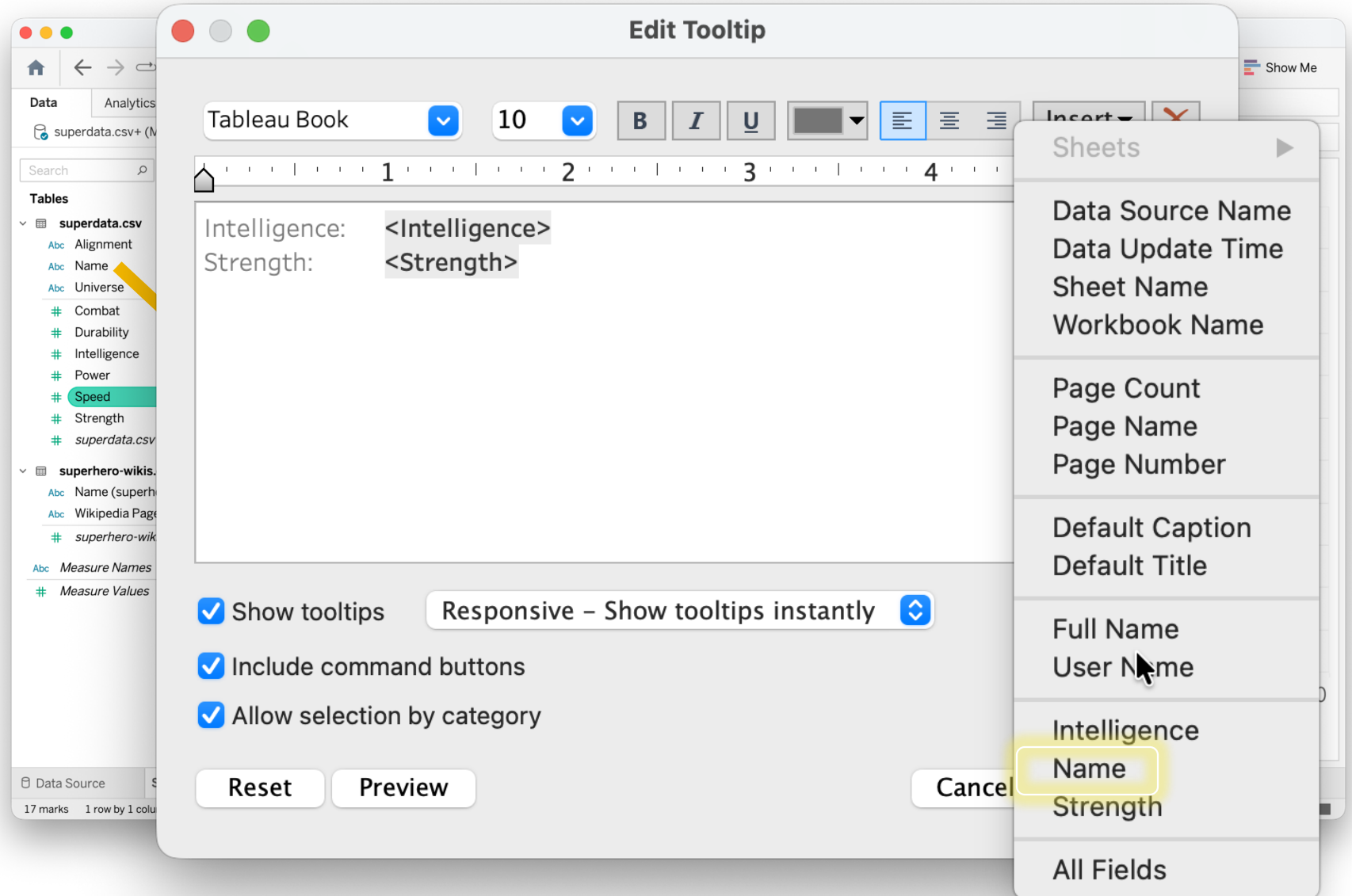
Finish sprucing



Better tooltips



Better tooltips



Better tooltips

The screenshot shows the 'Edit Tooltip' dialog box in Tableau. At the top, there's a title bar with standard macOS window controls (red, yellow, green buttons). Below the title bar is a toolbar with a dropdown menu set to 'Tableau Book', a font size dropdown set to '15', and buttons for bold (B), italic (I), underline (U), a color picker (black), bulleted list, numbered list, and indent list. To the right of these are an 'Insert' dropdown and a close button (X). Below the toolbar is a horizontal ruler with markers from 1 to 5. The main area is a text editor containing the following text: '<Name>' (highlighted in red), 'Intelligence: <Intelligence>' (with '<Intelligence>' highlighted in grey), and 'Strength: <Strength>' (with '<Strength>' highlighted in grey). A mouse cursor is positioned at the end of the third line. At the bottom of the dialog, there are three checked checkboxes: 'Show tooltips', 'Include command buttons', and 'Allow selection by category'. To the right of these is a dropdown menu set to 'On Hover - Show tooltips on hover'. At the very bottom are four buttons: 'Reset', 'Preview', 'Cancel', and 'OK'.

Tableau Book 15 B I U [Color Picker] [Bulleted List] [Numbered List] [Indent List] Insert X

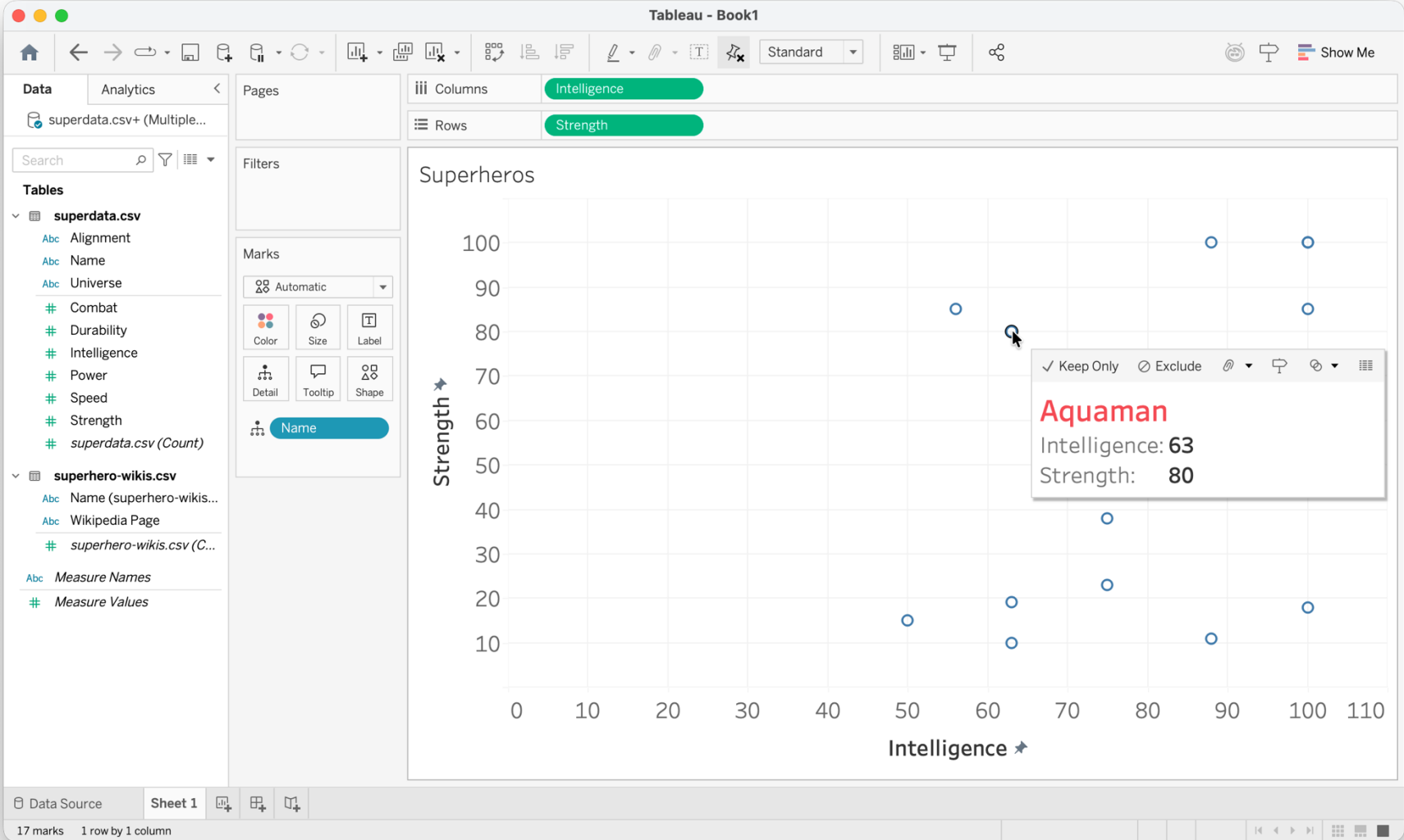
1 2 3 4 5

<Name>
Intelligence: <Intelligence>
Strength: <Strength>

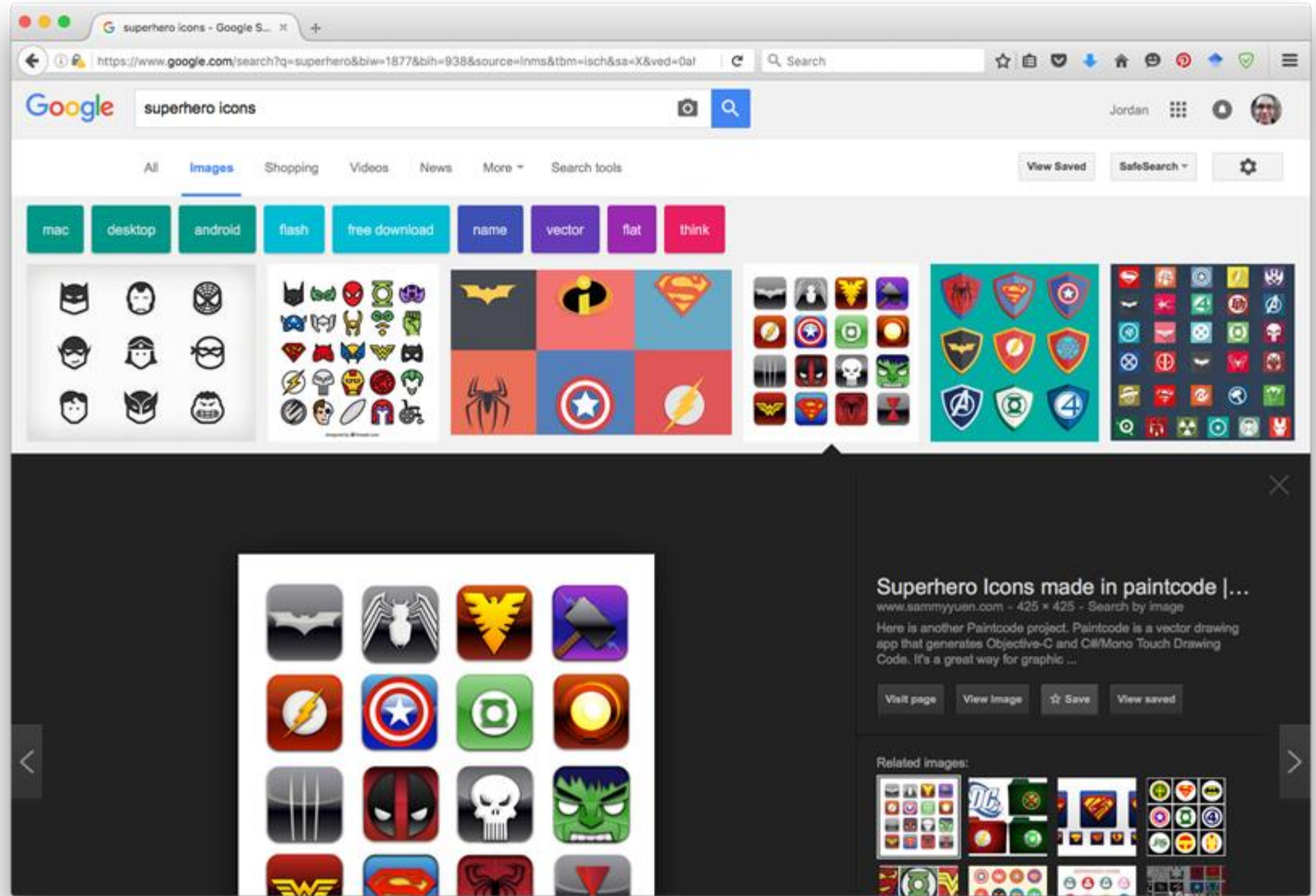
☒ Show tooltips On Hover – Show tooltips on hover
☒ Include command buttons
☒ Allow selection by category

Reset Preview Cancel OK

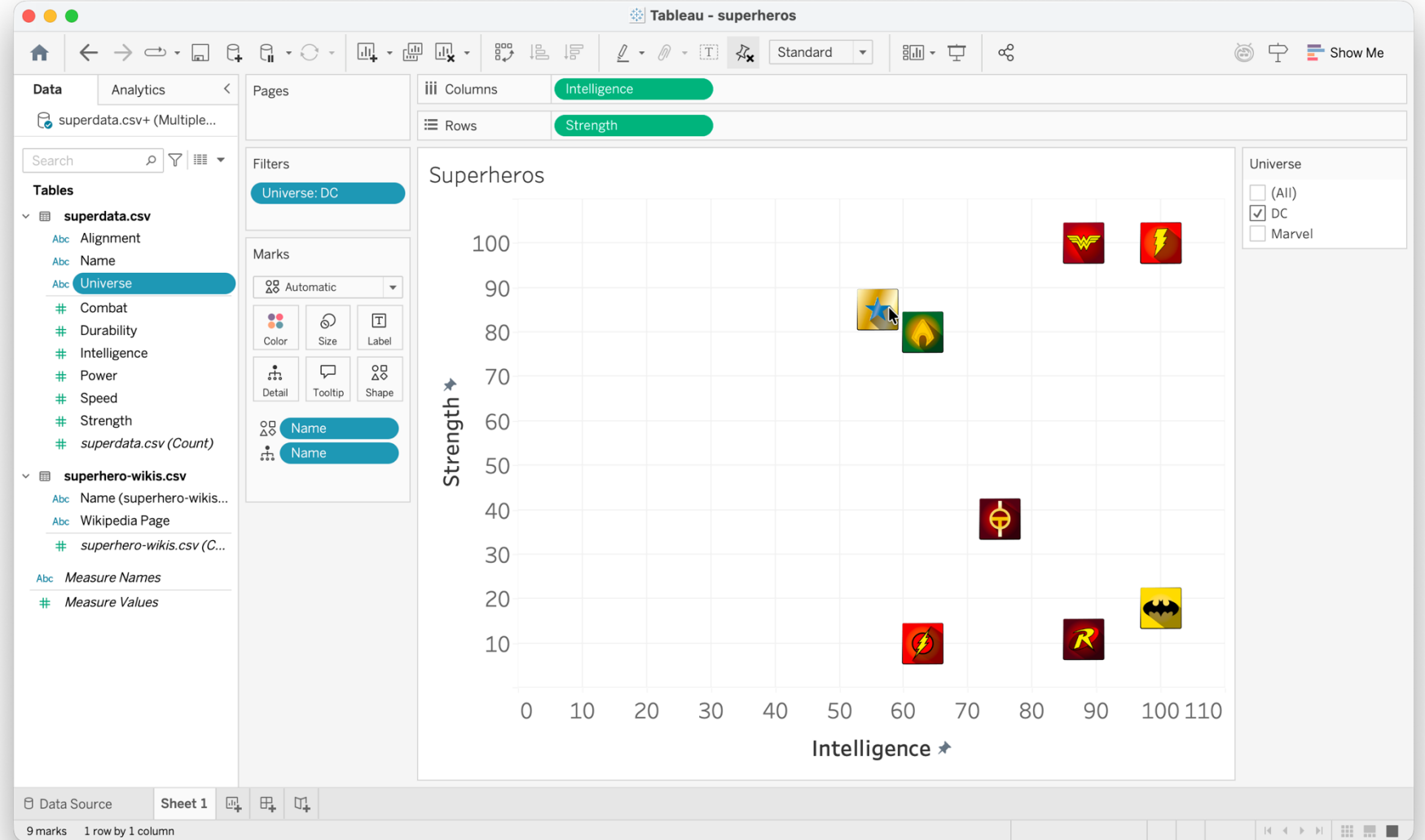
Better tooltips



Adding custom shapes



Working with filters



Changing a filter's scope

The screenshot shows the Tableau Desktop interface with a scatter plot titled 'Superheros'. The plot displays superhero icons on a grid with 'Intelligence' on the x-axis (ranging from 50 to 110) and 'Strength' on the y-axis. A filter for 'Universe' is applied, showing only DC and Marvel characters. The 'Edit Filter...' context menu is open, and the 'Apply to Worksheets' option is highlighted. A secondary menu is also visible, showing options for the filter's scope: 'All Using Related Data Sources', 'Using This Data Source', and 'Only This Worksheet'.

Tableau - superheros

Data | Analytics | Pages | Columns: Intelligence | Rows: Strength

Filters: Universe: DC

Tables

- superdata.csv+** (Multiple...)
 - Alignment
 - Name
 - Universe
 - Combat
 - Durability
 - Intelligence
 - Power
 - Speed
 - Strength
 - superdata.csv (Count)
- superhero-wikis.csv**
 - Name (superhero-wikis...)
 - Wikipedia Page
 - superhero-wikis.csv (C...
- Measure Names
- Measure Values

Marks

- Automatic
- Color
- Size
- Detail
- Tooltip
- Name
- Name

Universe

- (All)
- ☒ DC
- ☐ Marvel

Edit Filter...

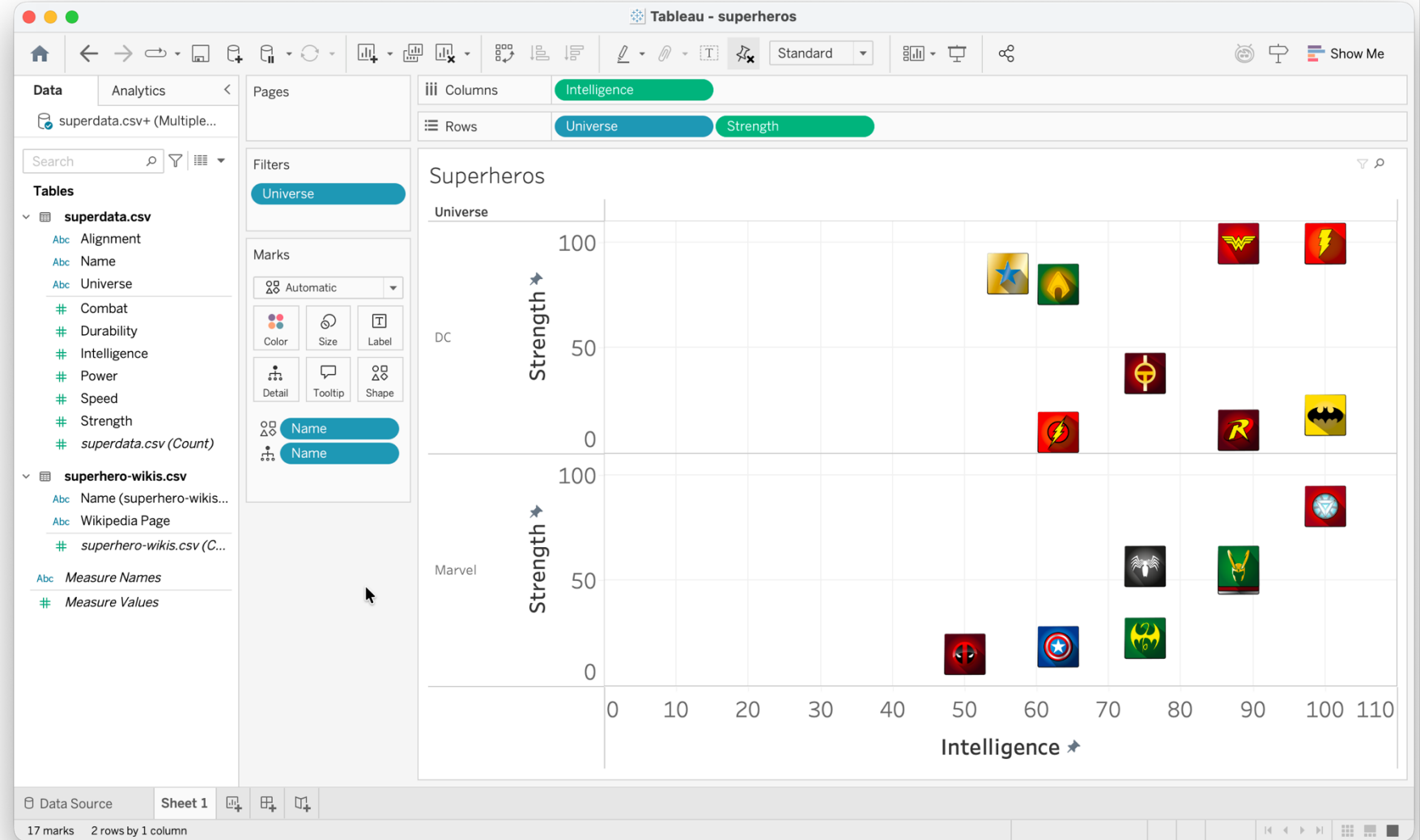
- ✓ Show Filter
- Clear Filter
- Add to Context
- Apply to Worksheets**
- Create Set...
- Find in Data pane
- ✓ Dimension
- Attribute
- Measure
- Remove

All Using Related Data Sources

- ☒ Using This Data Source
- ☒ Only This Worksheet

9 marks 1 row by 1 column

Implicit filtering (side- by-side comparison)



Adding
background
images

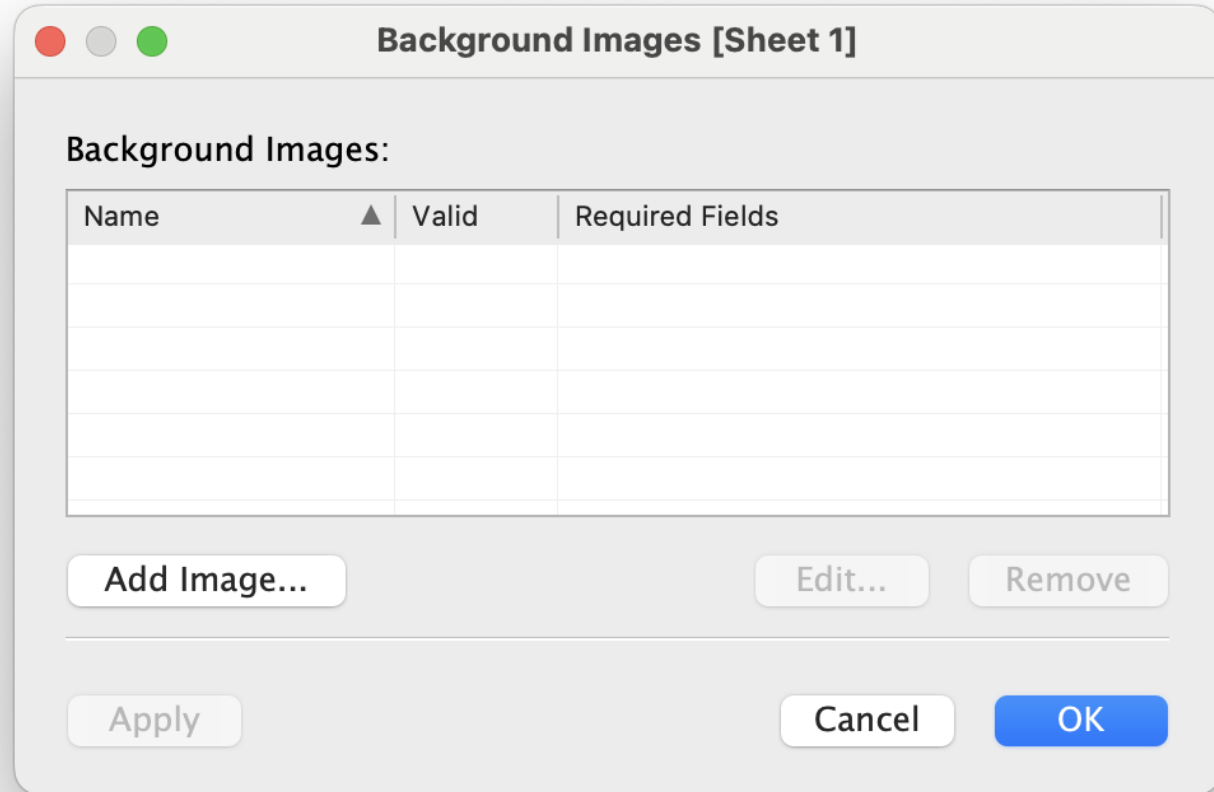


Adding background images

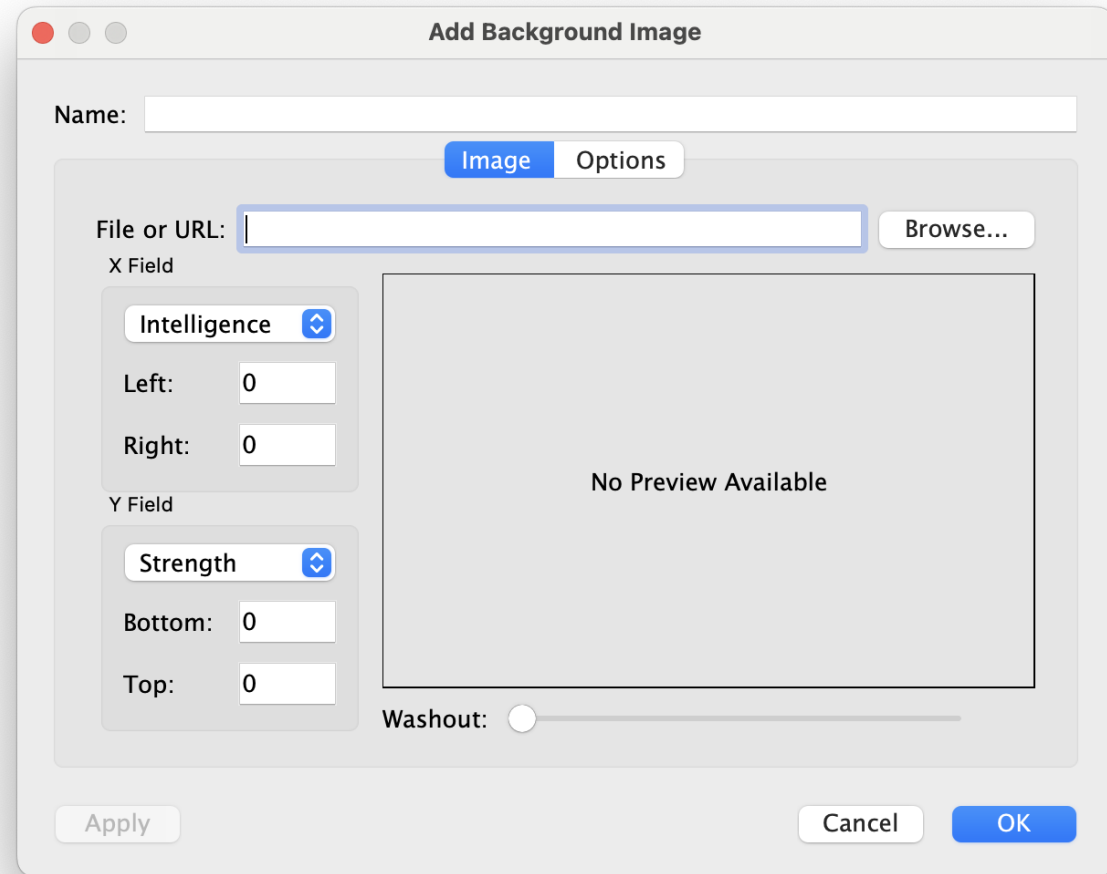
The screenshot shows the Tableau Desktop interface. The main view is a scatter plot titled 'Strength' on the y-axis and 'Intelligence' on the x-axis. The plot displays various superhero icons as data points. A context menu is open over the plot, showing options: 'Background Maps', 'Background Images', 'Geocoding', 'Map Legend', 'Map Layers...', and 'Map Options...'. The 'Background Images' option is highlighted, and a submenu is visible showing 'superdata.csv+ (Multiple Connections)...'. The left sidebar shows the 'Format Workbook' pane with 'Tableau Book' selected. The right sidebar shows the 'Universe' pane with 'All', 'DC', and 'Marvel' selected. The bottom status bar indicates '17 marks' and '2 rows by 1 column'.

Universe	Name	Intelligence	Strength
Marvel	Aquaman	65	10
Marvel	Batman	60	10
Marvel	Booster Gold	65	10
Marvel	Captain America	60	10
Marvel	Captain Marvel	65	10
Marvel	Deadpool	60	10
Marvel	Flash	65	10
Marvel	Iron Fist	60	10
Marvel	Iron Man	65	10
Marvel	Loki	60	10
Marvel	Red Tornado	65	10
Marvel	Robin	60	10
Marvel	Spider-Man	65	10
Marvel	Superman	60	10
Marvel	Venom	65	10
Marvel	Wonder Woman	60	10
Marvel	X-Man	65	10

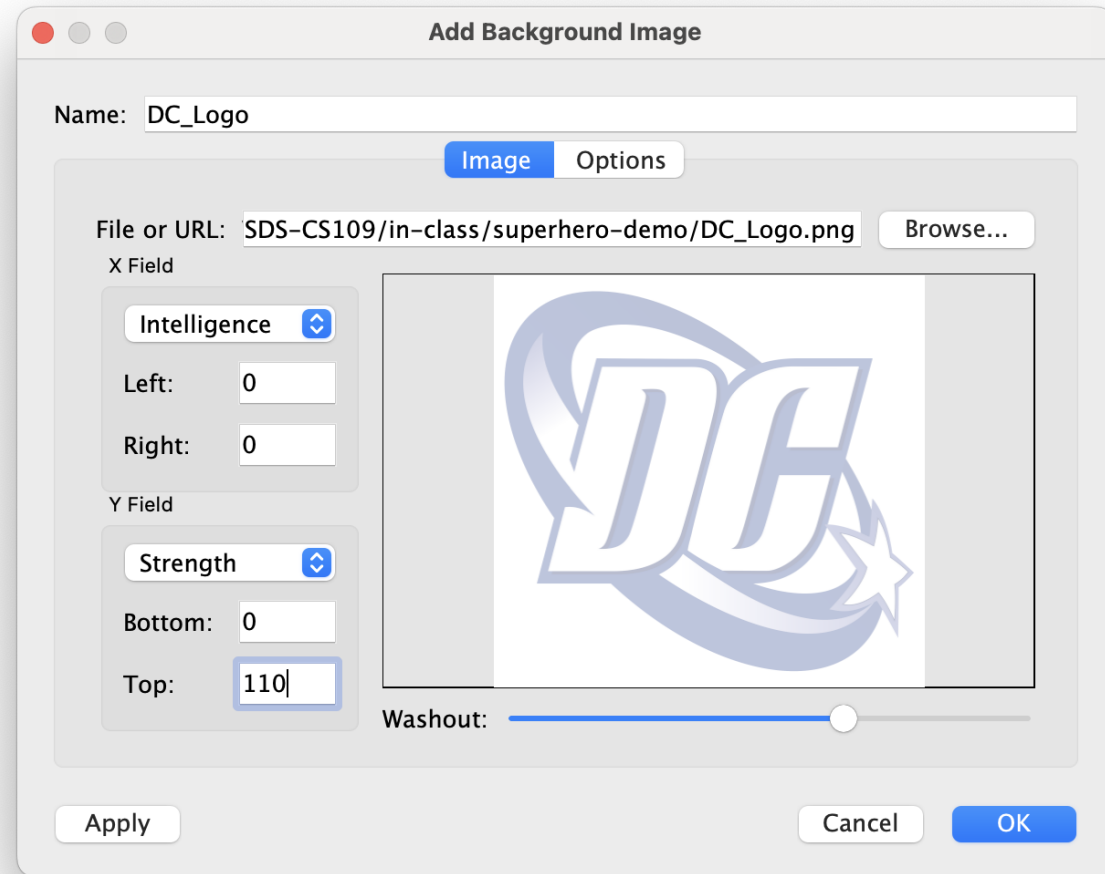
Adding background images



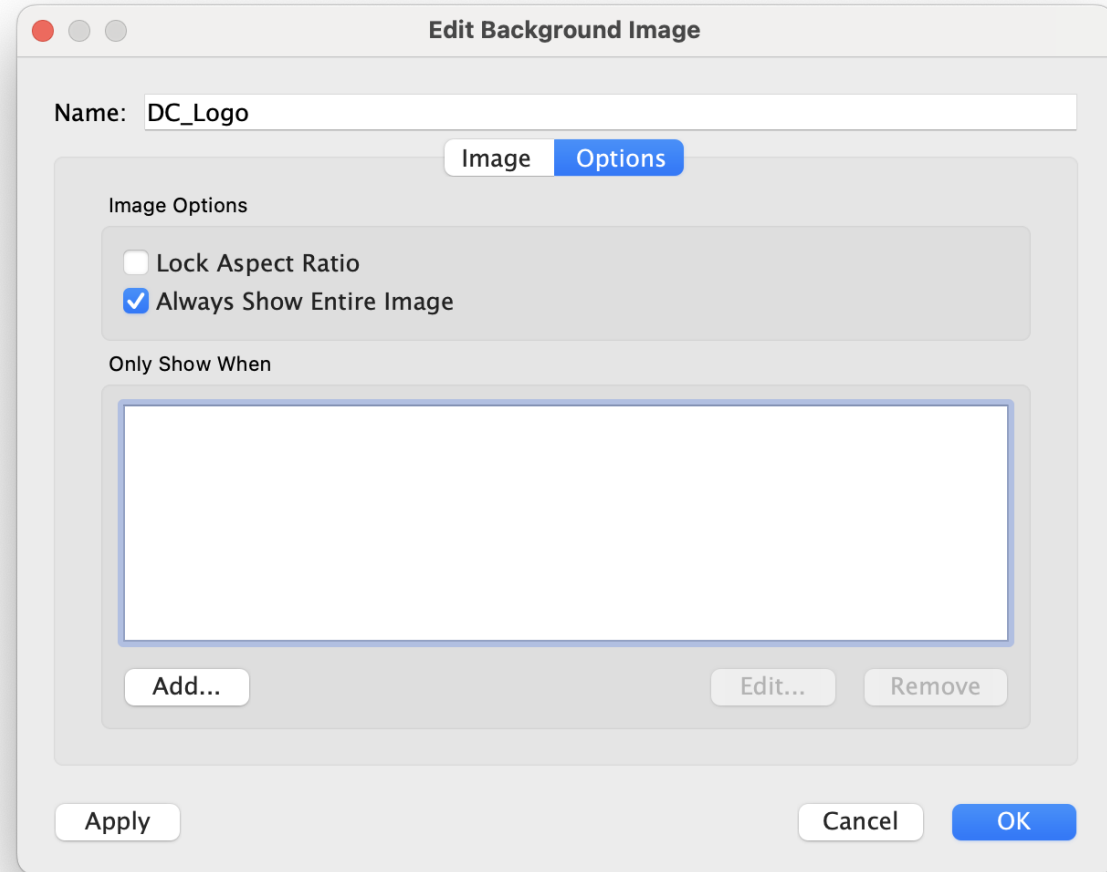
Adding background images



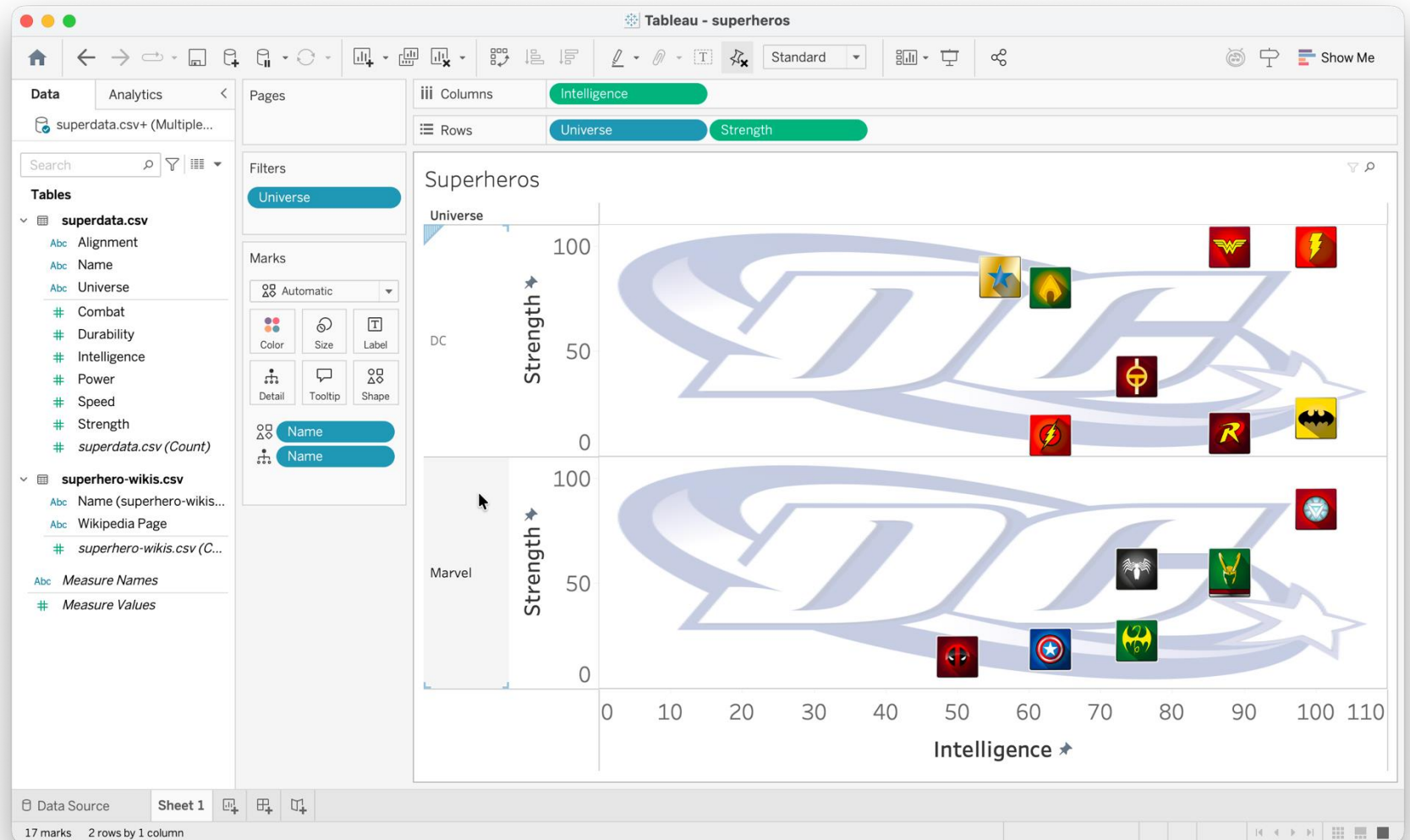
Adding background images



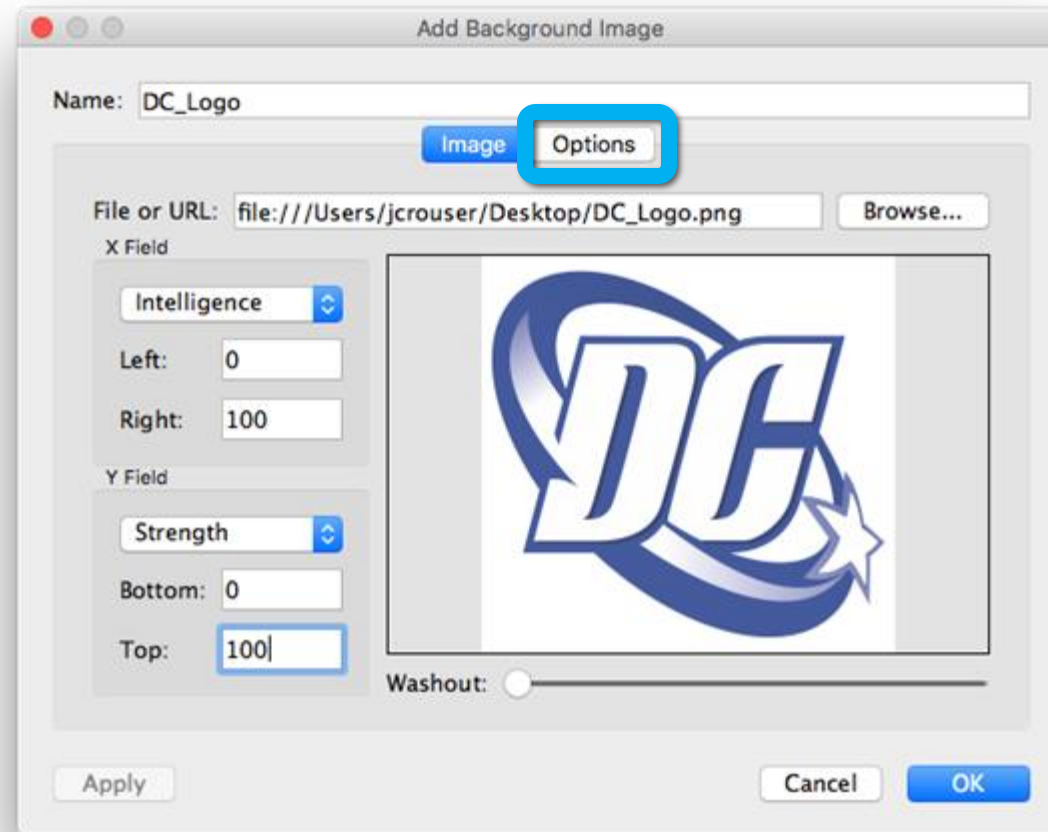
Adding background images



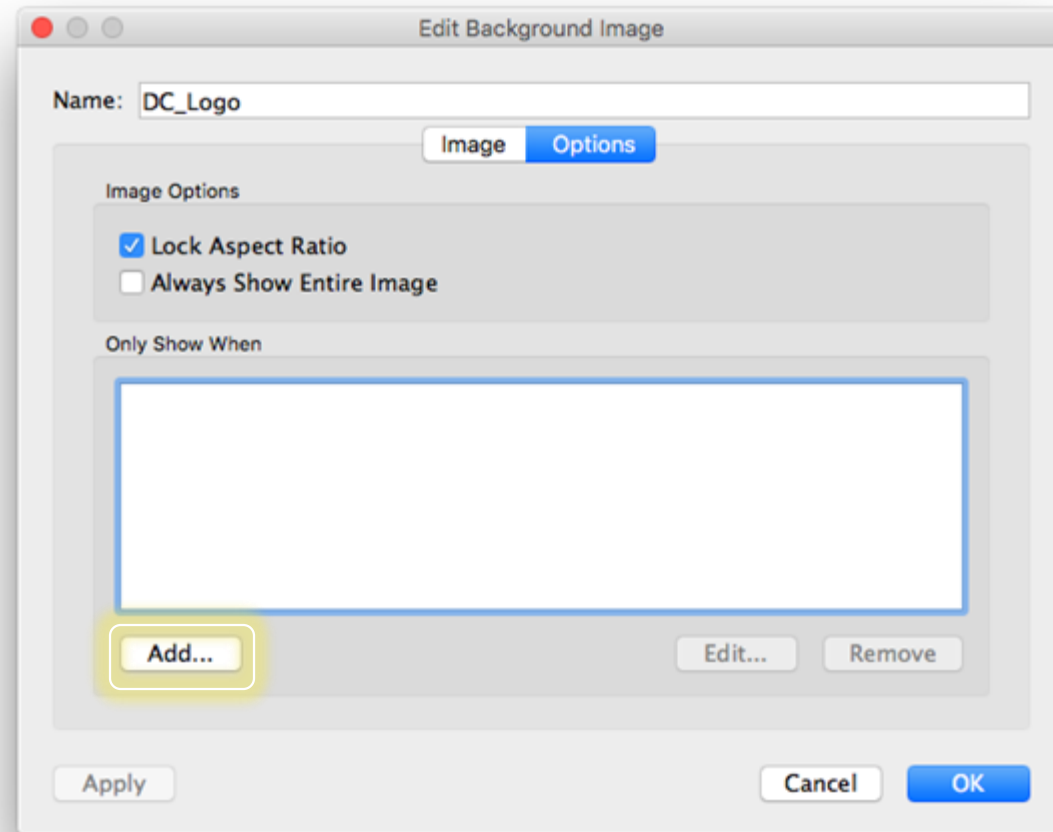
Adding background images



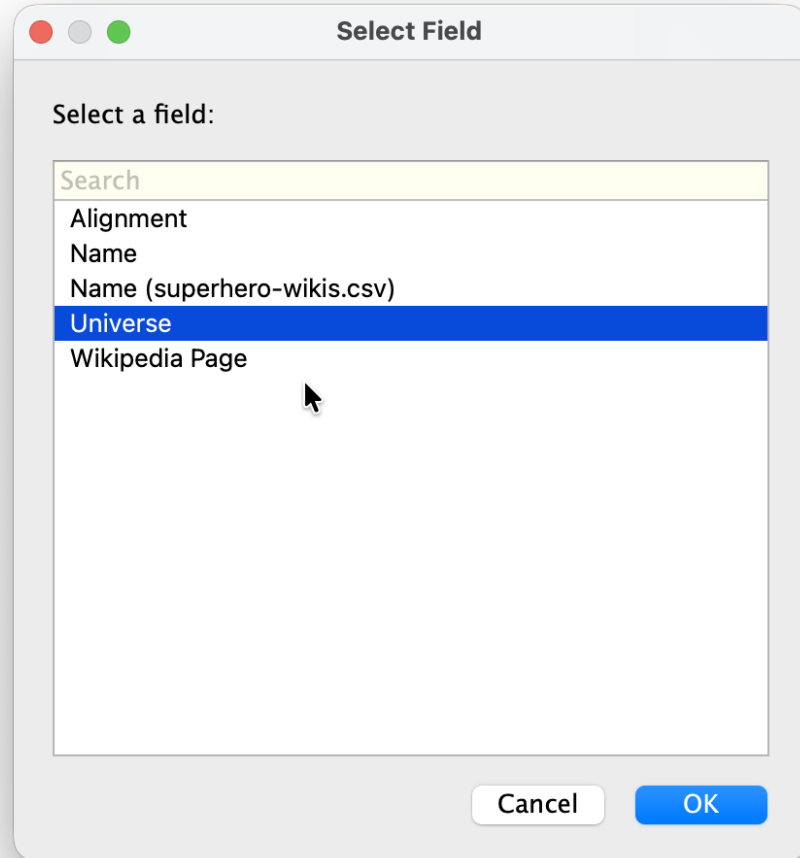
Adding background images



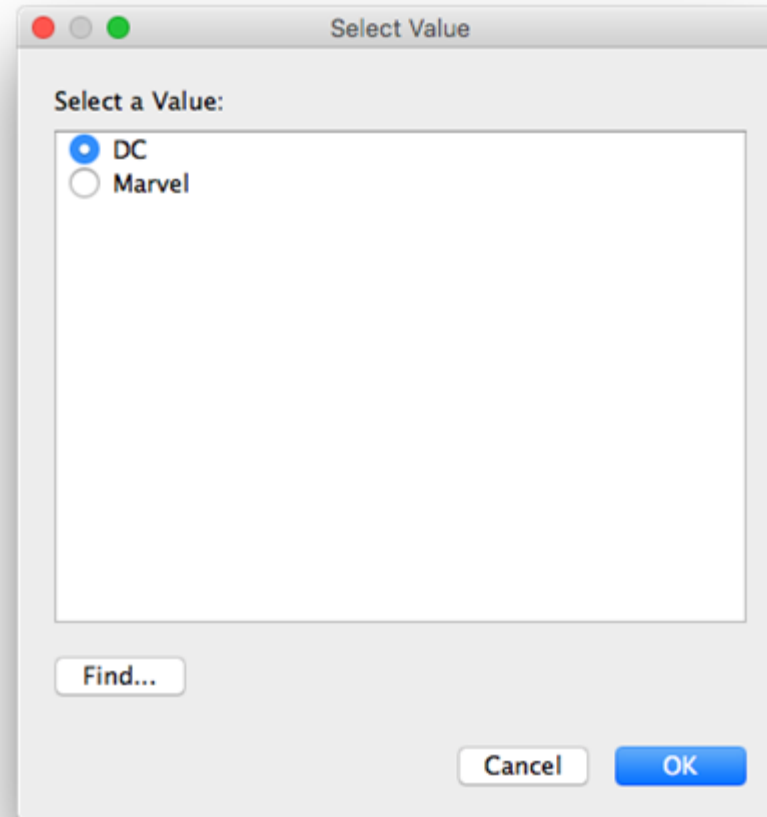
Adding background images



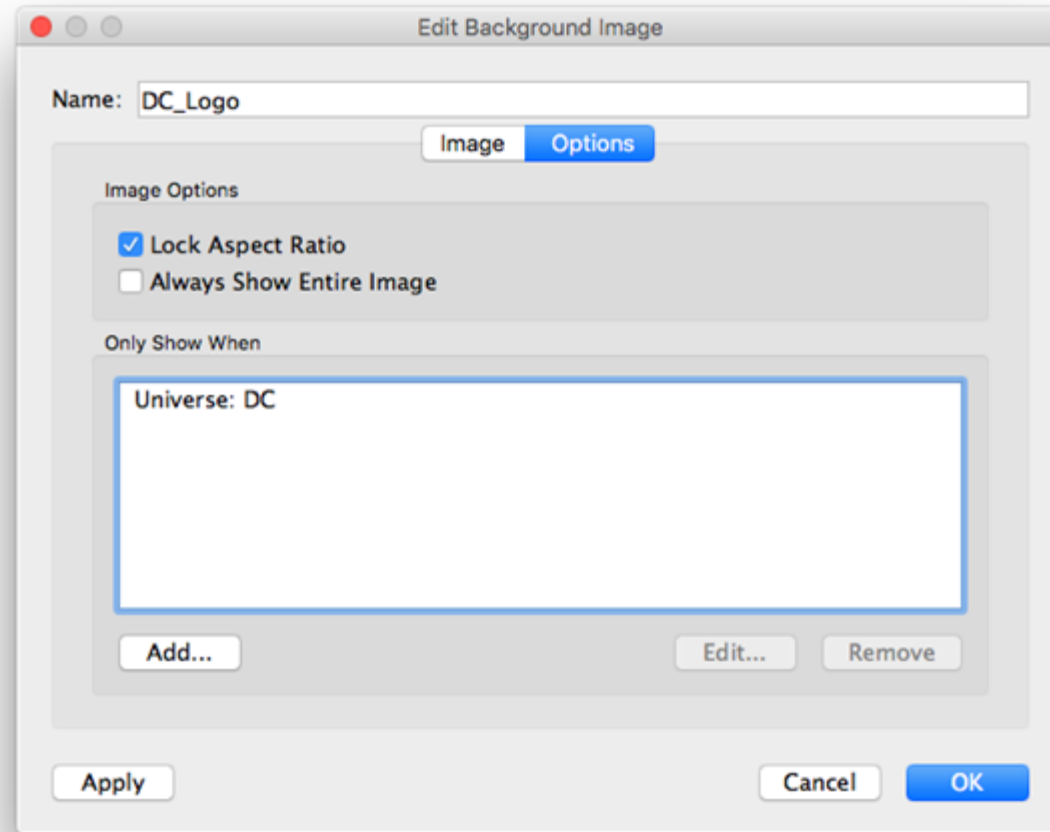
Adding background images



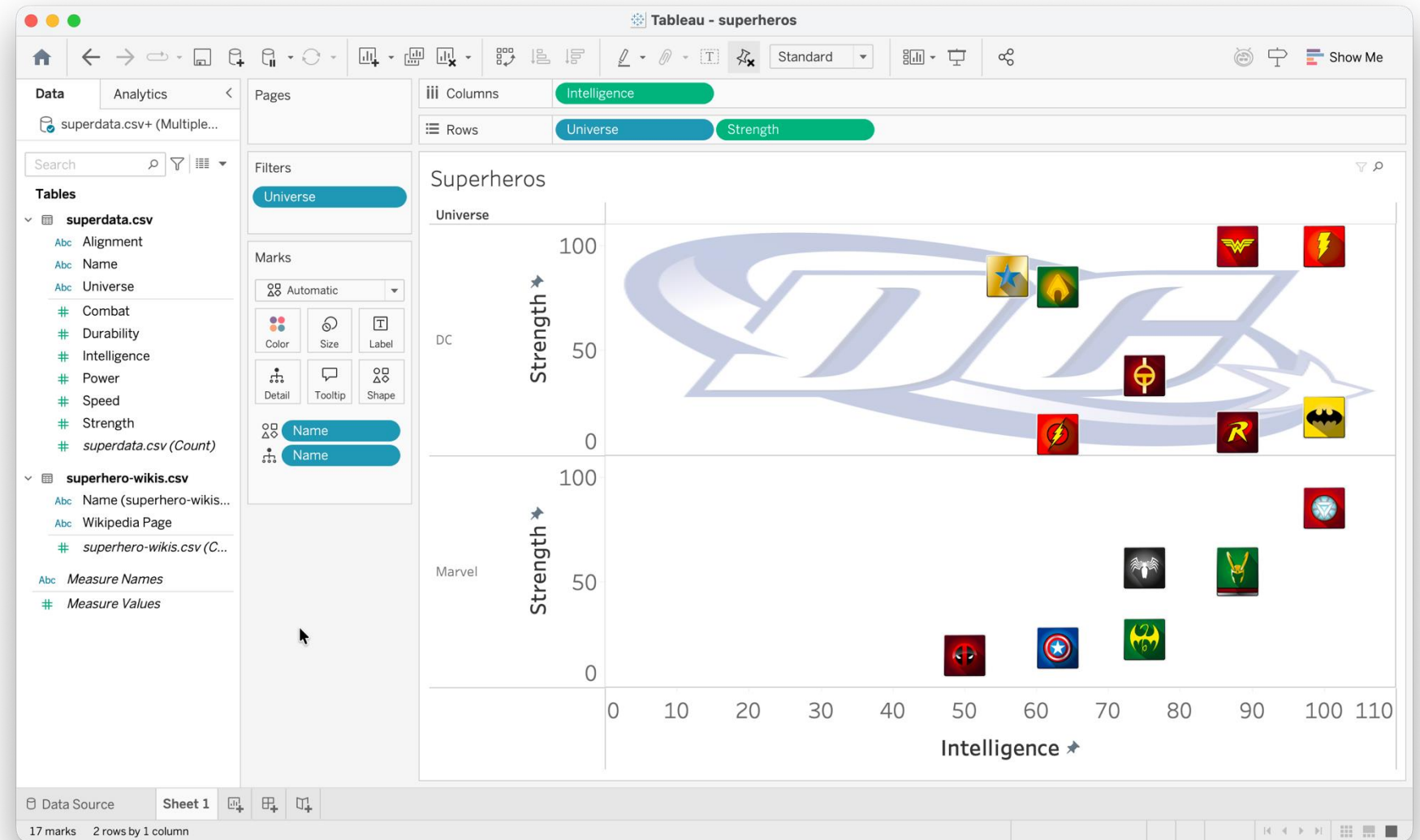
Adding background images



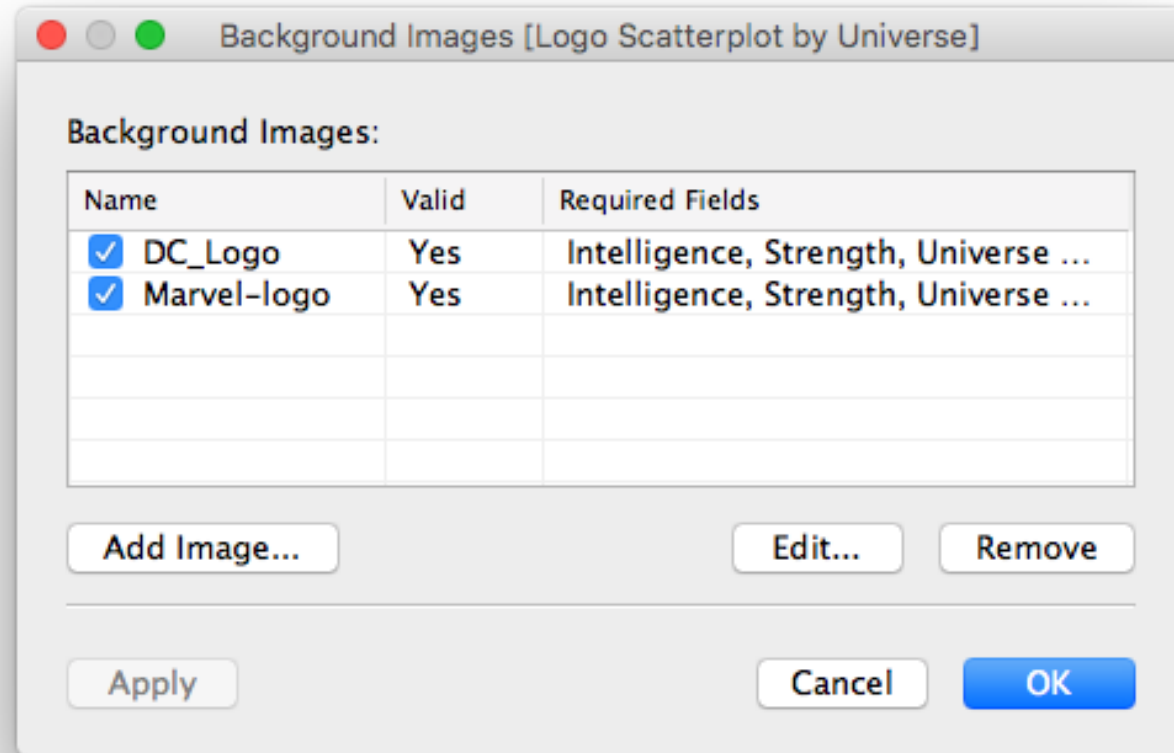
Adding background images



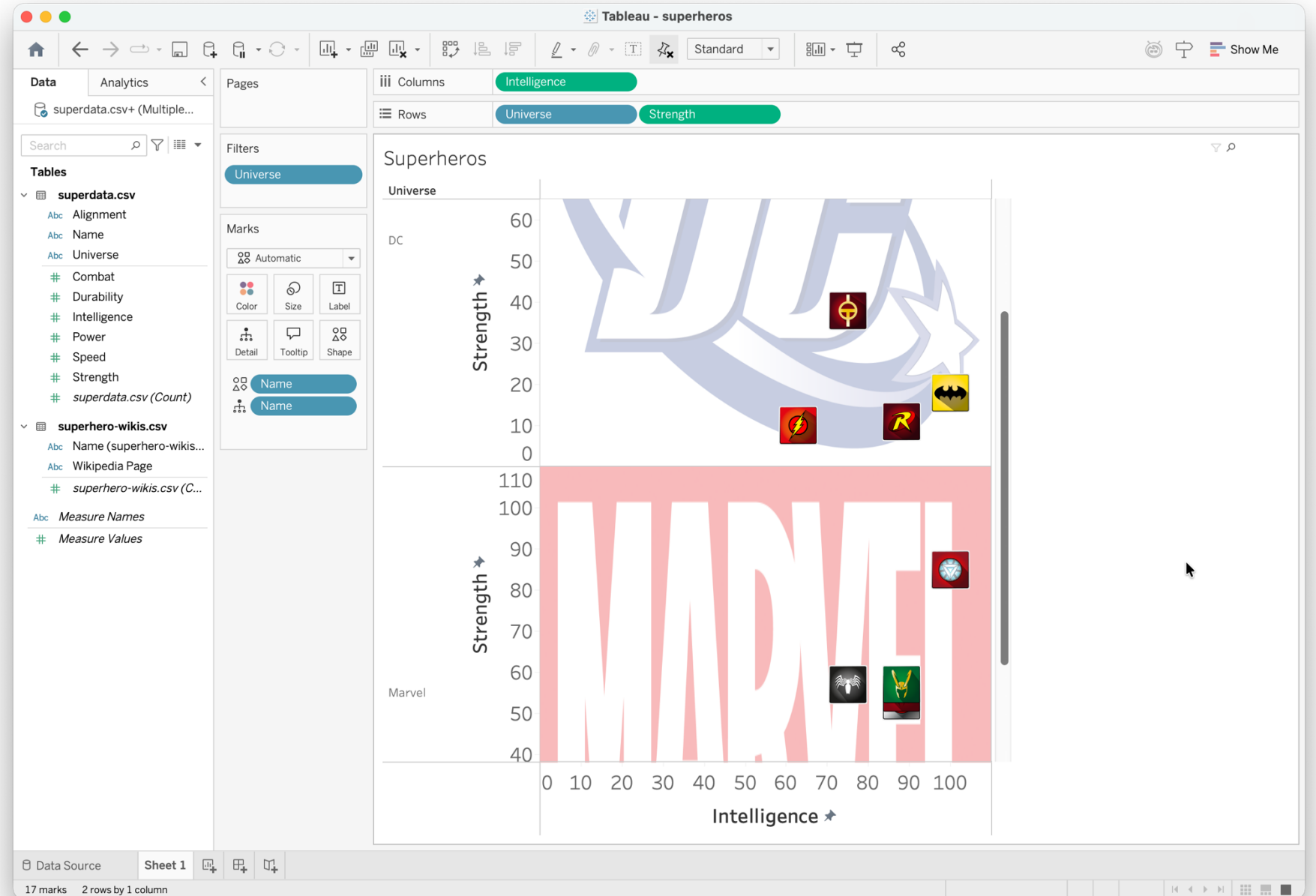
Adding background images



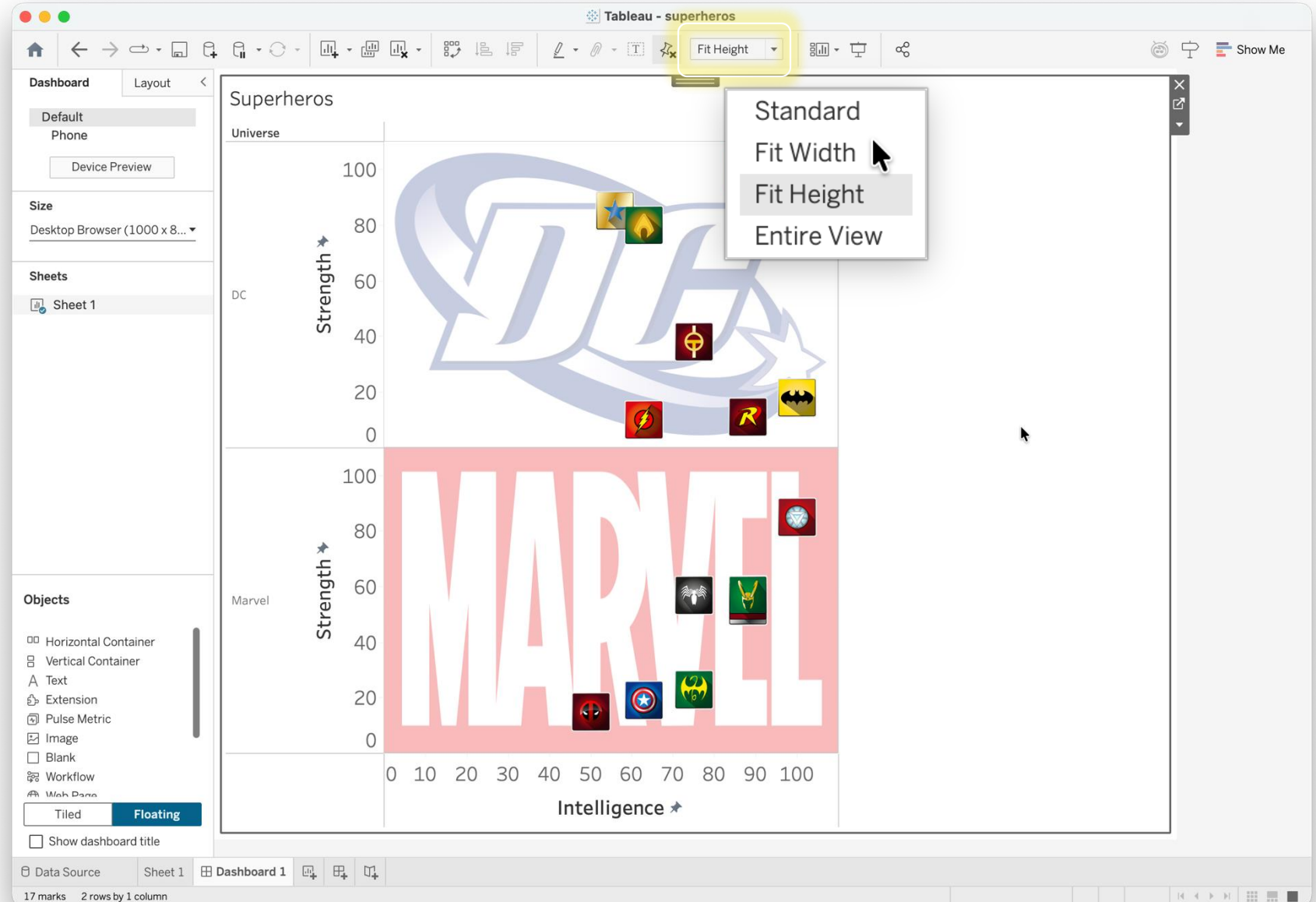
Adding background images



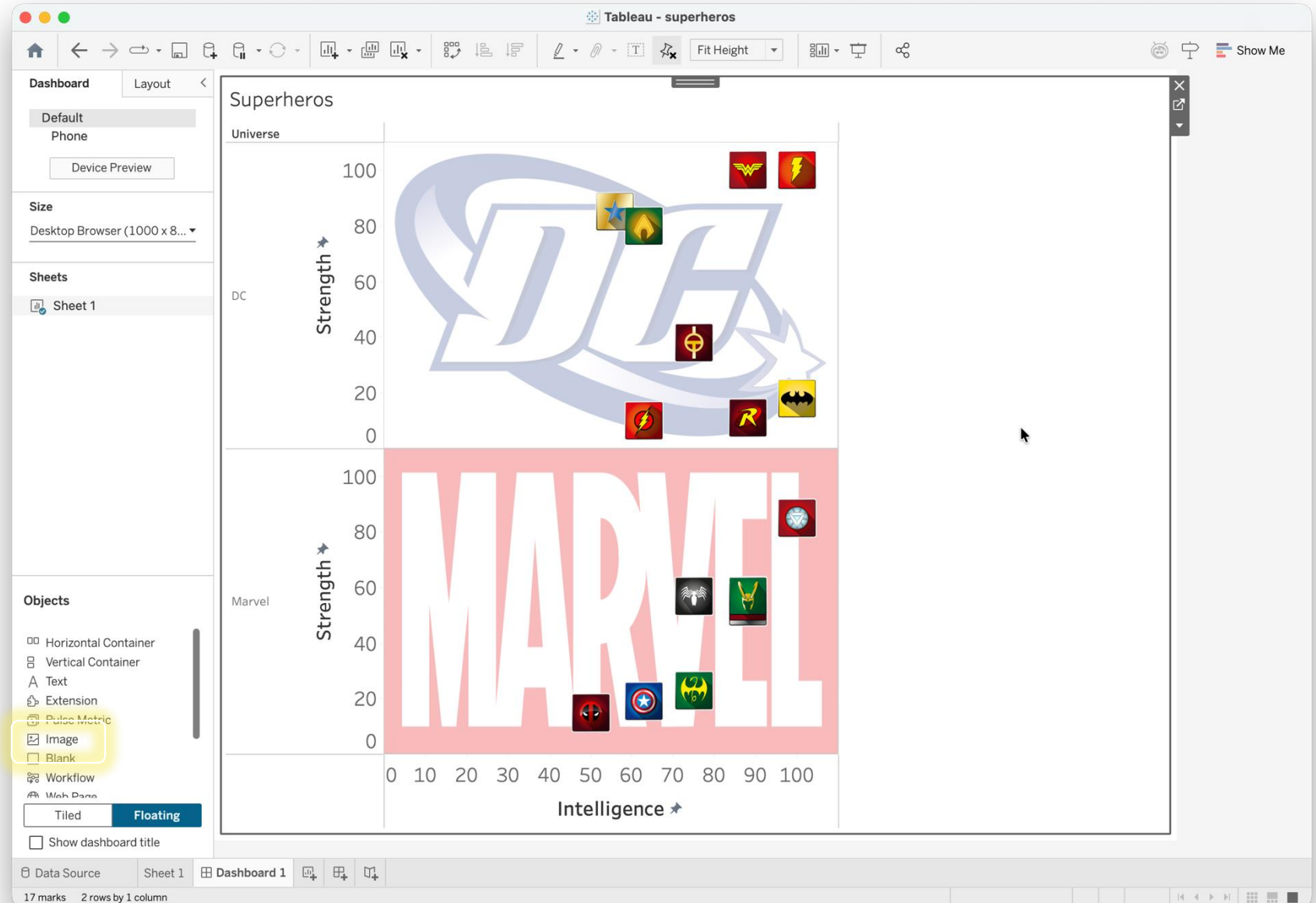
Adding background images



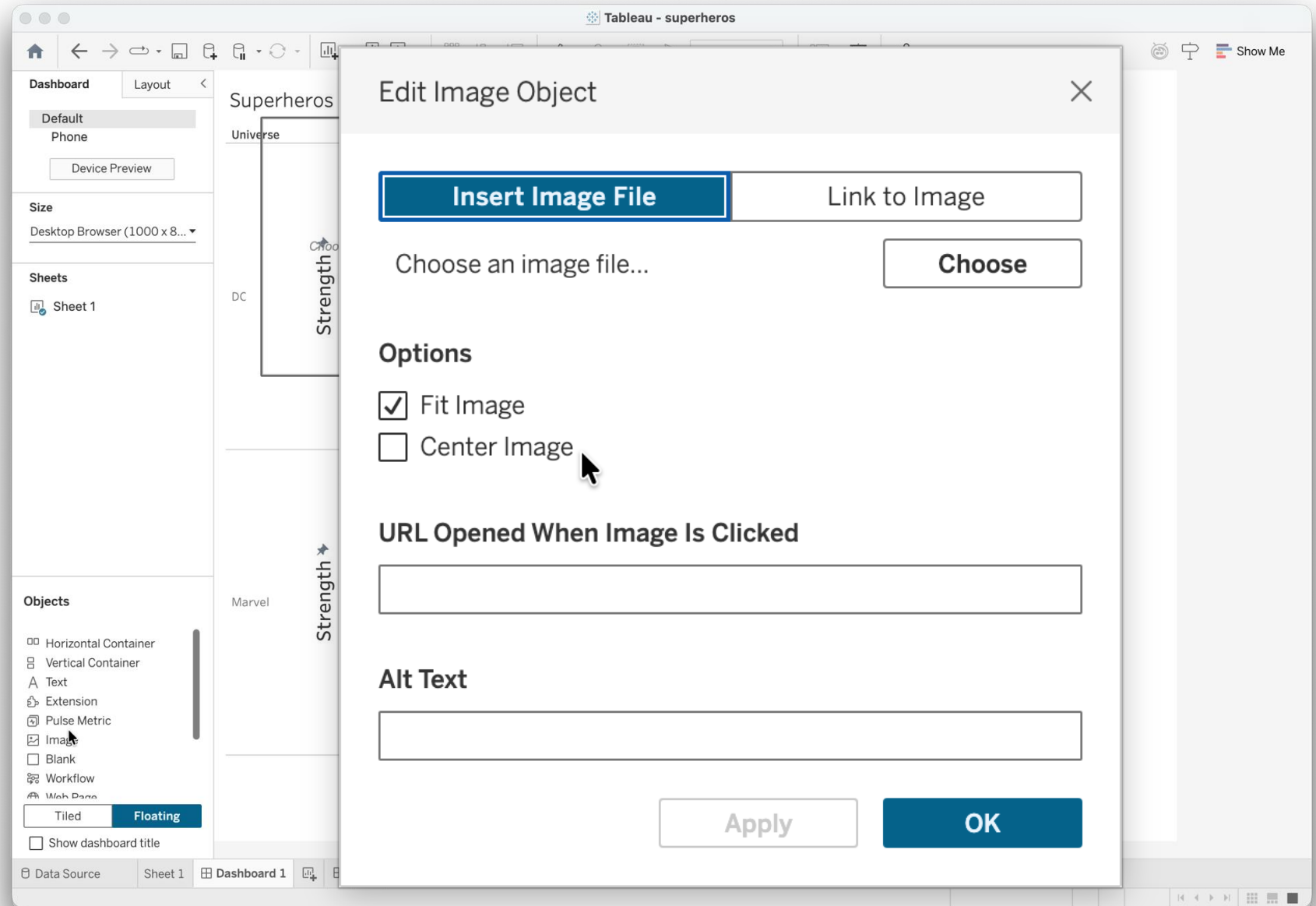
Customizing a dashboard



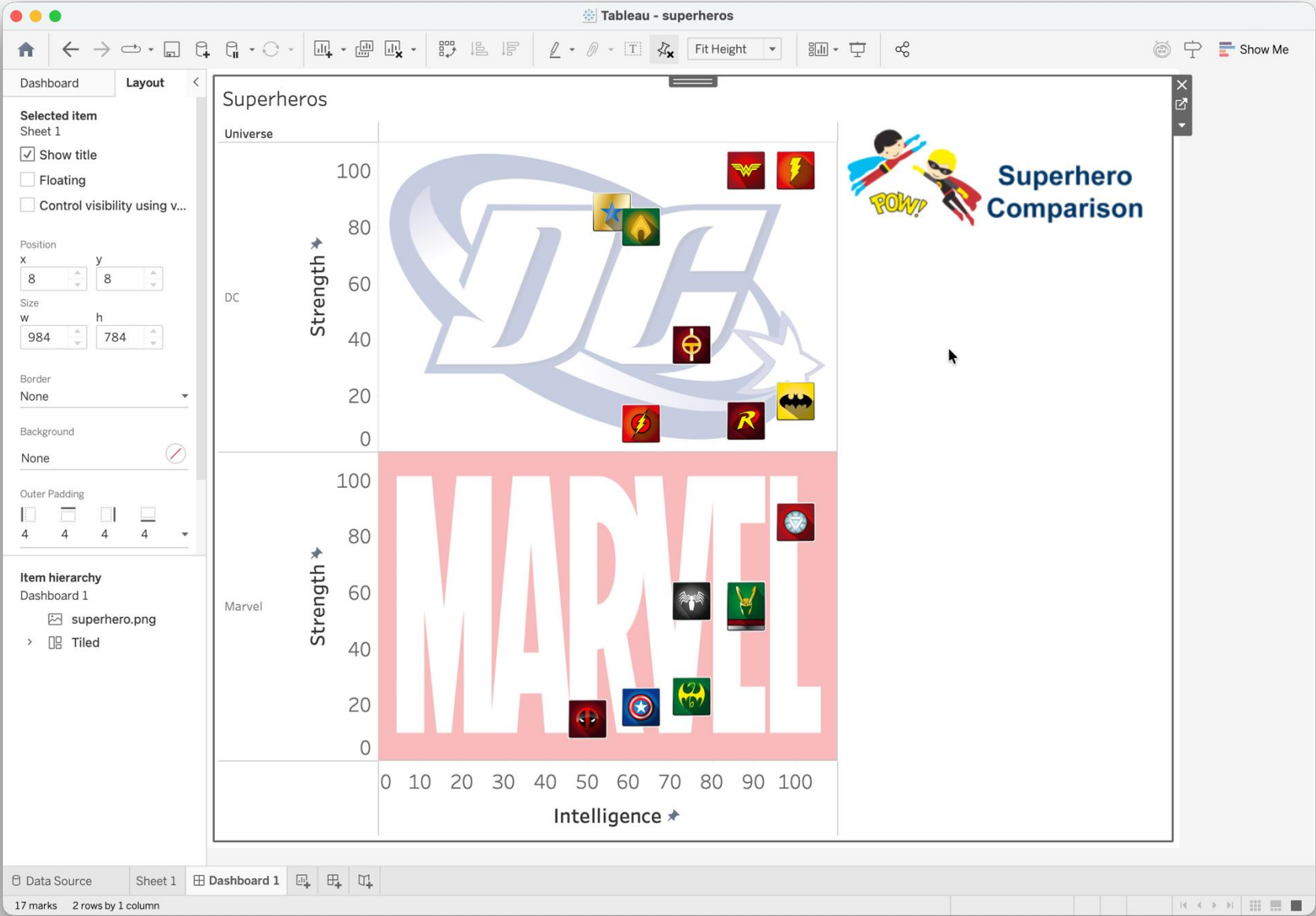
Customizing a dashboard



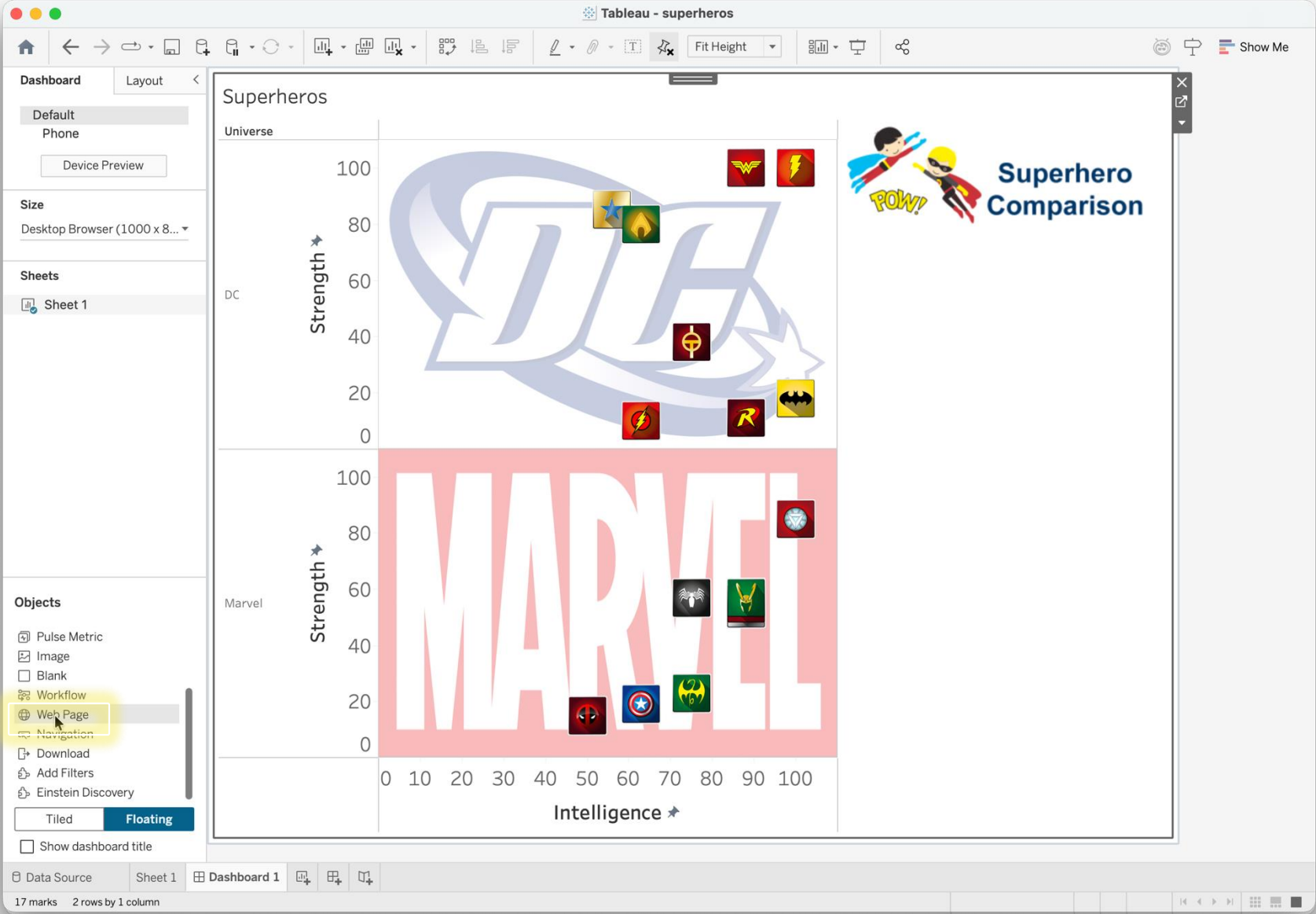
Customizing a dashboard



Customizing a dashboard



Load external data on click



Load external
data on click

The image shows a Tableau Desktop window titled "Tableau - superheros" in the background. Overlaid on top is a web browser window titled "Login | Salesforce" showing the Salesforce login page. The browser's address bar displays the URL: `login.salesforce.com/?startURL=%2Fsetup%2Fsecur%2FRemoteAccessAuthorizationPage.apexp%3Fs...`. The Salesforce login page features the Salesforce logo, a login form with fields for "Username" and "Password", a "Log In" button, a "Remember me" checkbox, and links for "Forgot Your Password?" and "Use Custom Domain". Below the login form is a "Try for Free" button. To the right of the login form, the page says "Start your free trial. No credit card required, no software to install." and lists benefits of the trial, such as "Preloaded data or upload your own" and "Preconfigured processes, reports, and dashboards". A large blue button at the bottom right says "Start my free trial". The Tableau Desktop interface in the background includes a sidebar with "Objects" (Pulse M, Image, Blank, Workflow, Web Pa, Navigat, Downlo, Add Filt, Einstein Discovery) and a bottom pane with "Data Source", "Sheet 1", and "Dashboard 1". The "Intelligence" tab is also visible.

Tableau - superheros

Dashboard

Default

Phone

De

Size

Desktop B

Sheets

Sheet

Objects

Pulse M

Image

Blank

Workflo

Web Pa

Navigat

Downlo

Add Filt

Einstein Discovery

Tiled

Floating

Intelligence

Data Source

Sheet 1

Dashboard 1

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Login | Salesforce

login.salesforce.com/?startURL=%2Fsetup%2Fsecur%2FRemoteAccessAuthorizationPage.apexp%3Fs...

salesforce

Username

Password

Log In

☐ Remember me

[Forgot Your Password?](#) [Use Custom Domain](#)

Not a customer? [Try for Free](#)

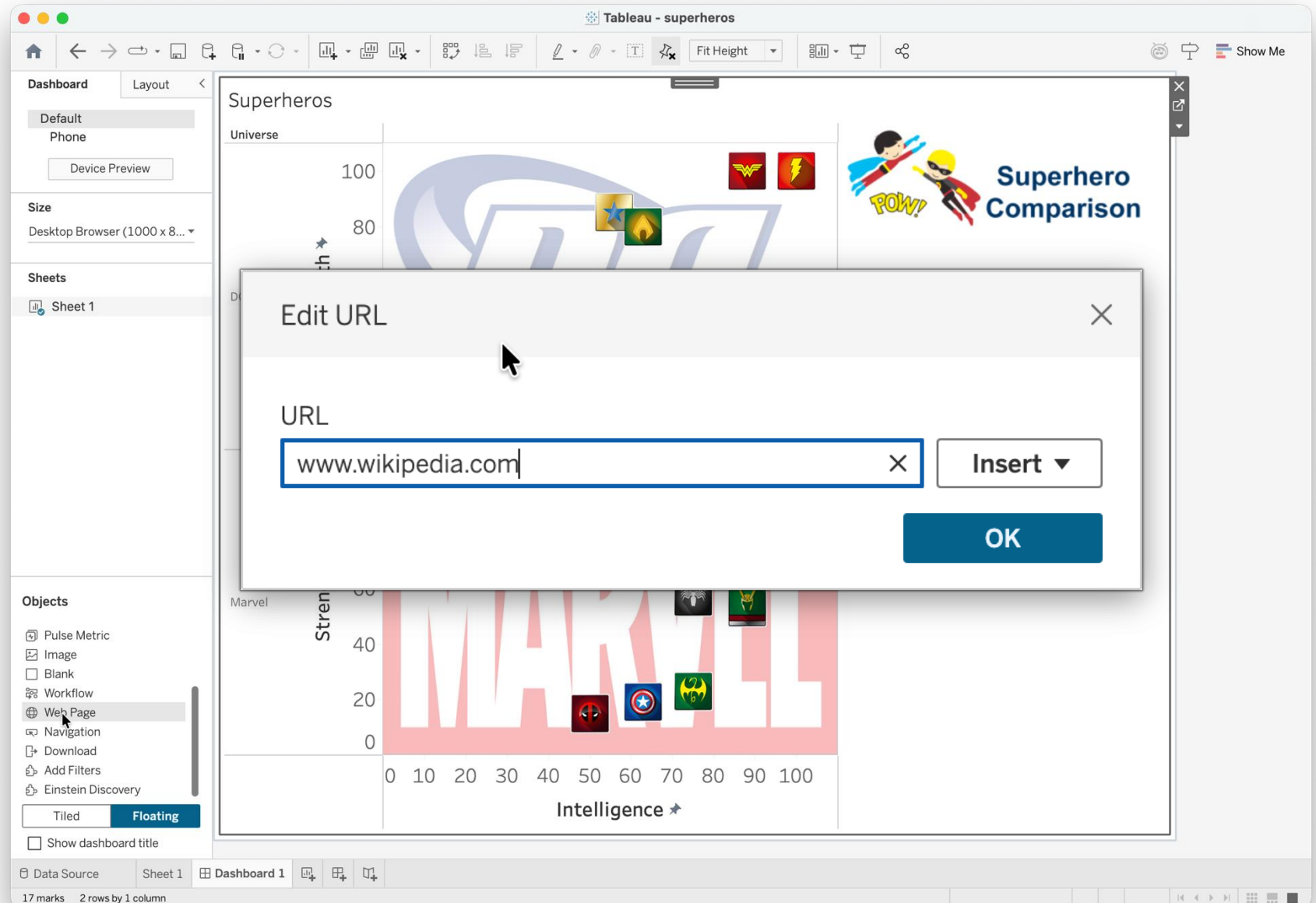
Start your free trial. No credit card required, no software to install.

With your trial, you get:

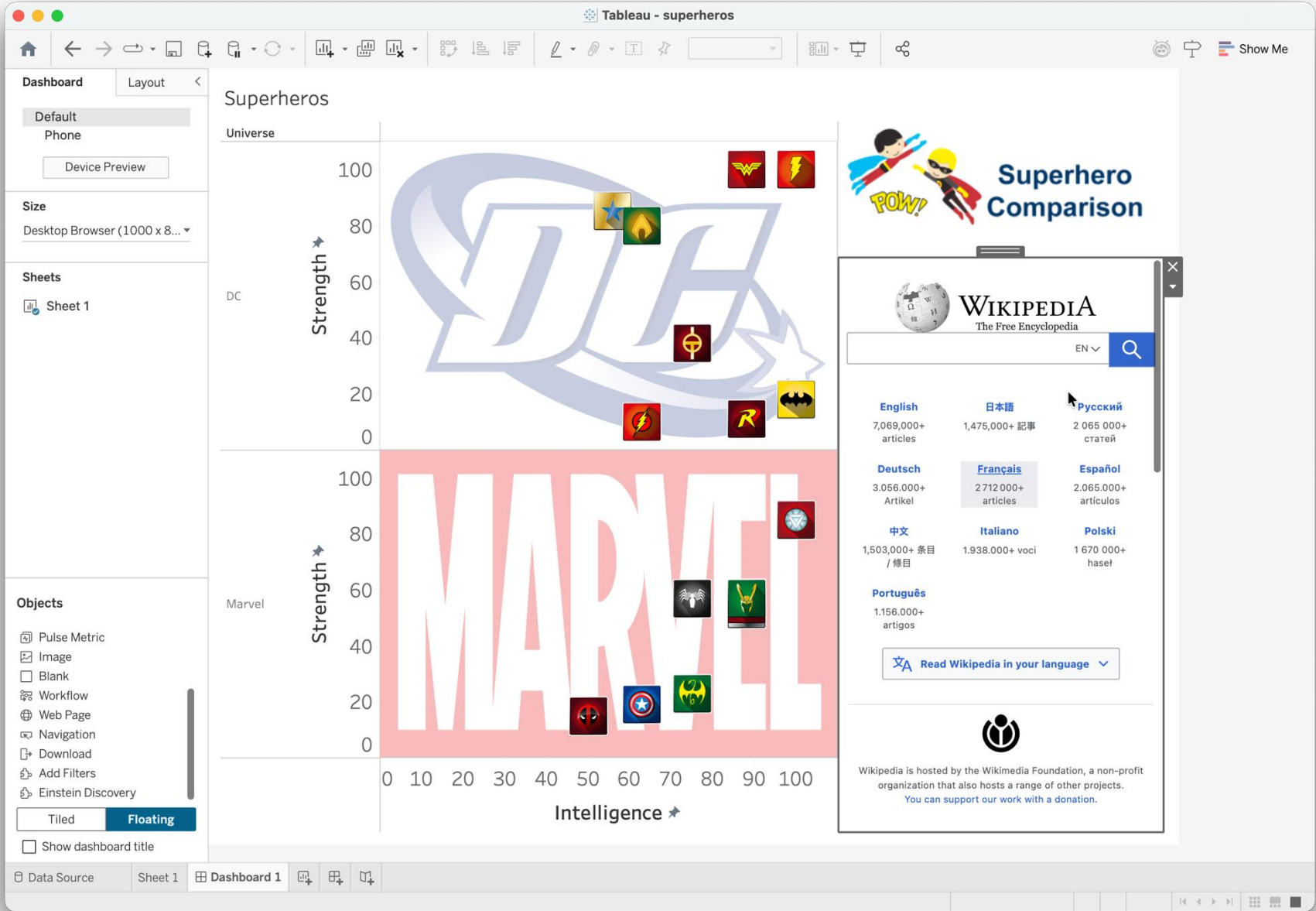
- ✓ Preloaded data or upload your own
- ✓ Preconfigured processes, reports, and dashboards
- ✓ Guided experiences for sales reps, leaders, and administrators
- ✓ Online training and live onboarding webinars

Start my free trial

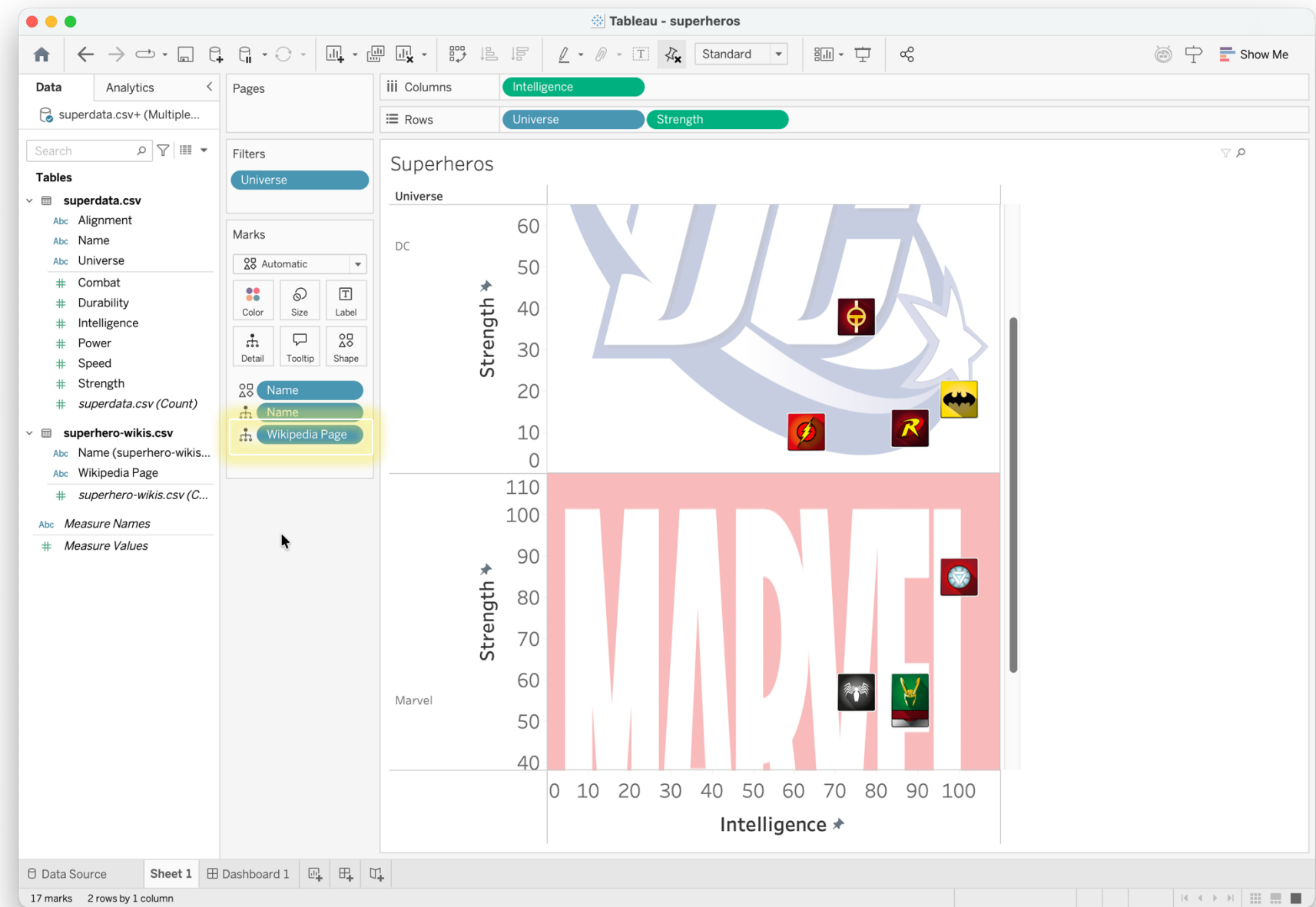
Load external
data on click



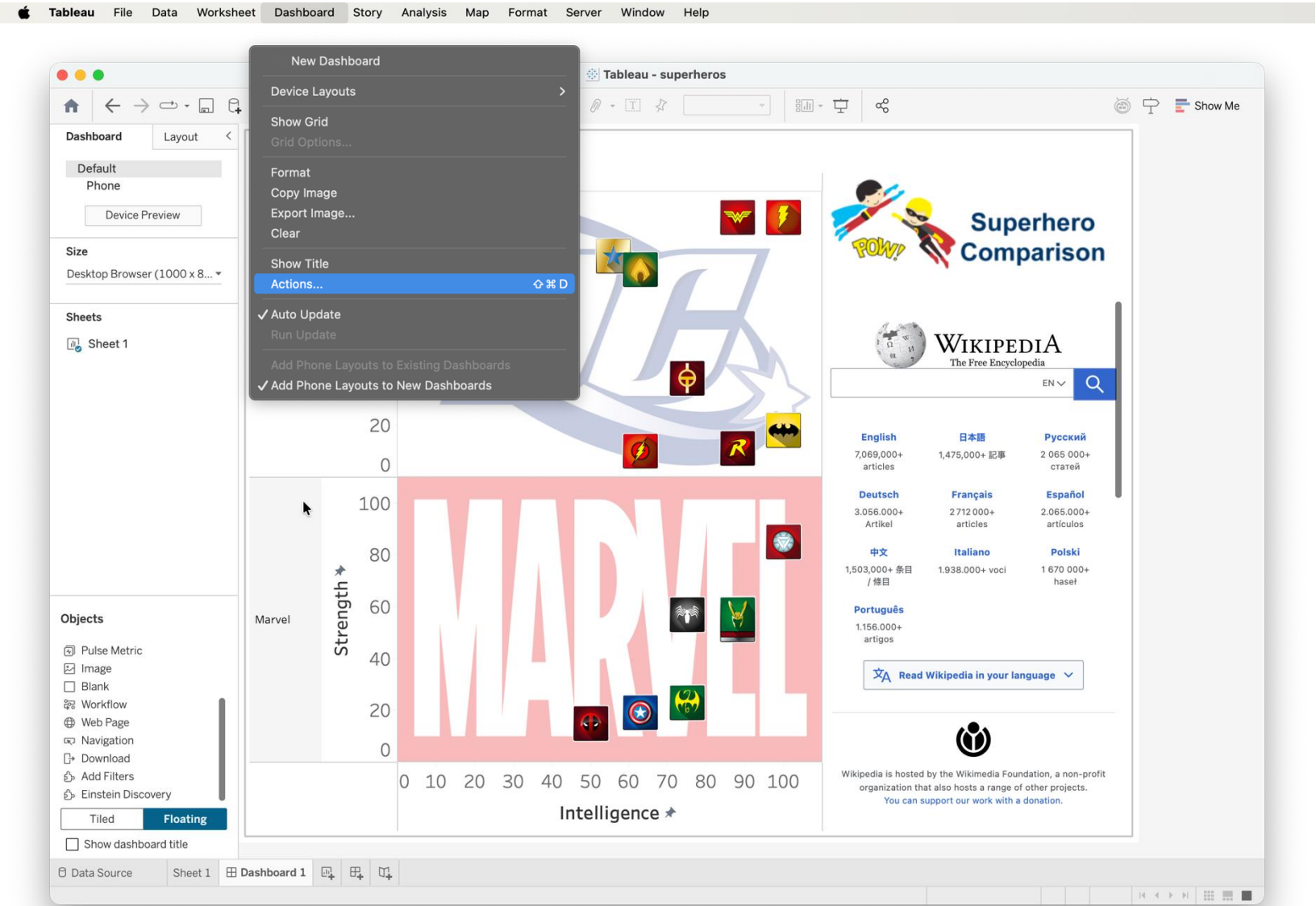
Load external data on click



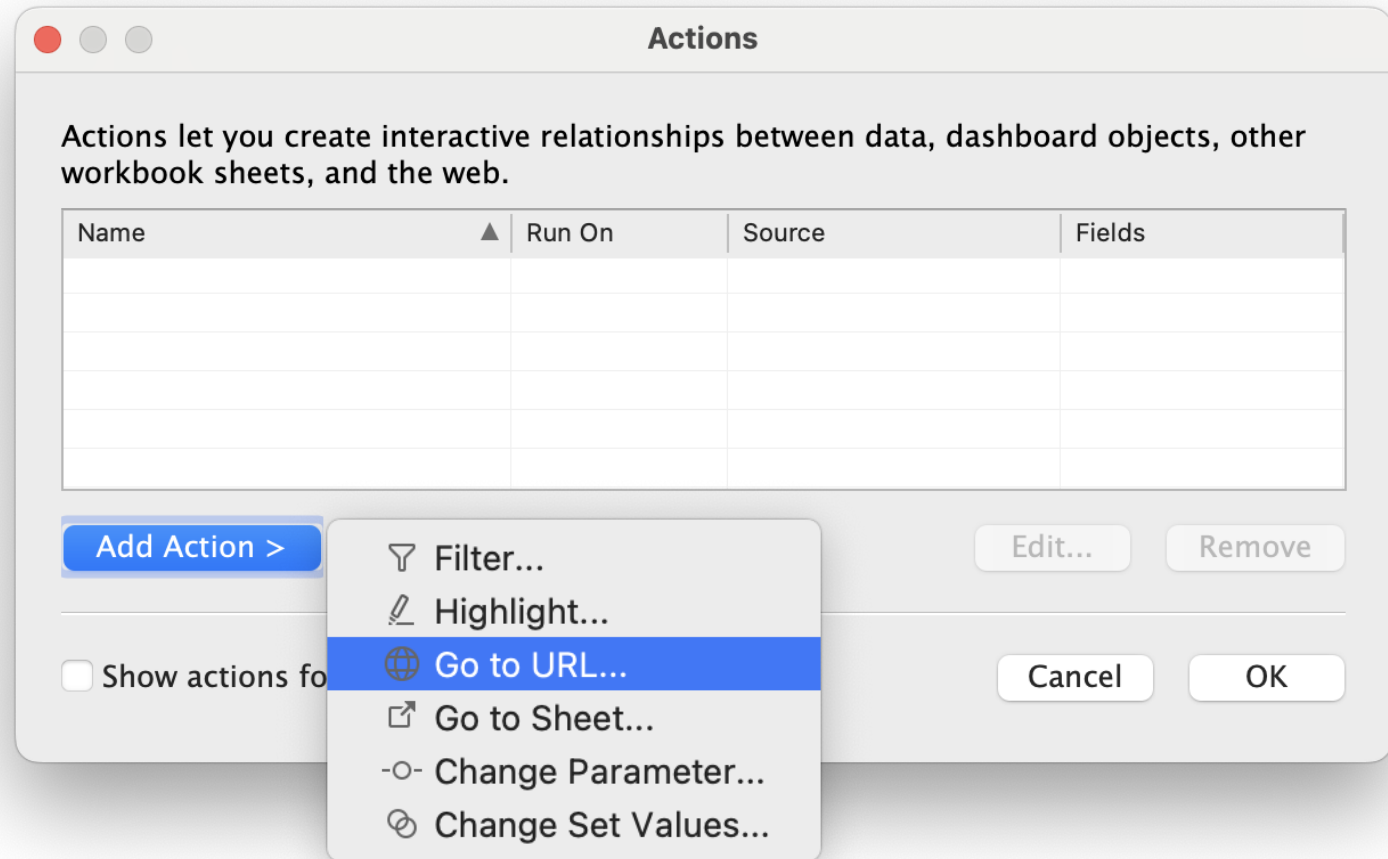
Load external data on click



Load external data on click



Load external data on click



Load external
data on click

Edit URL Action

✕

Name

Hyperlink1

Insert ▼

Source Sheets

Dashboard 1

▼

☒ Sheet 1

Run action on

☐ Hover

☐ Select

☒ Menu

URL Target

☐ New Tab if No Web Page Object Exists

☐ New Browser Tab

☒ Web Page Object

www.wikipedia.com

▼

URL

<Wikipedia Page>

Insert ▼

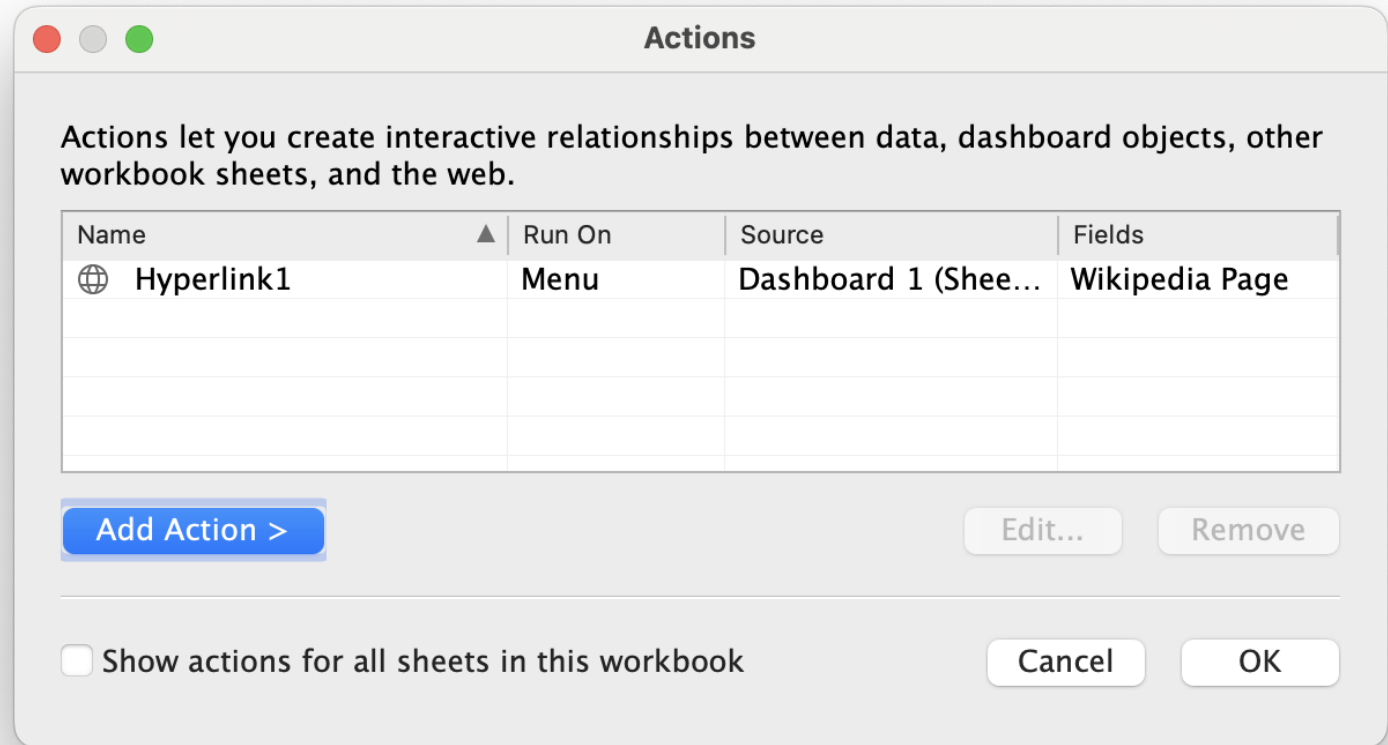
<https://en.m.wikipedia.org/wiki/Aquaman>

> Data Values

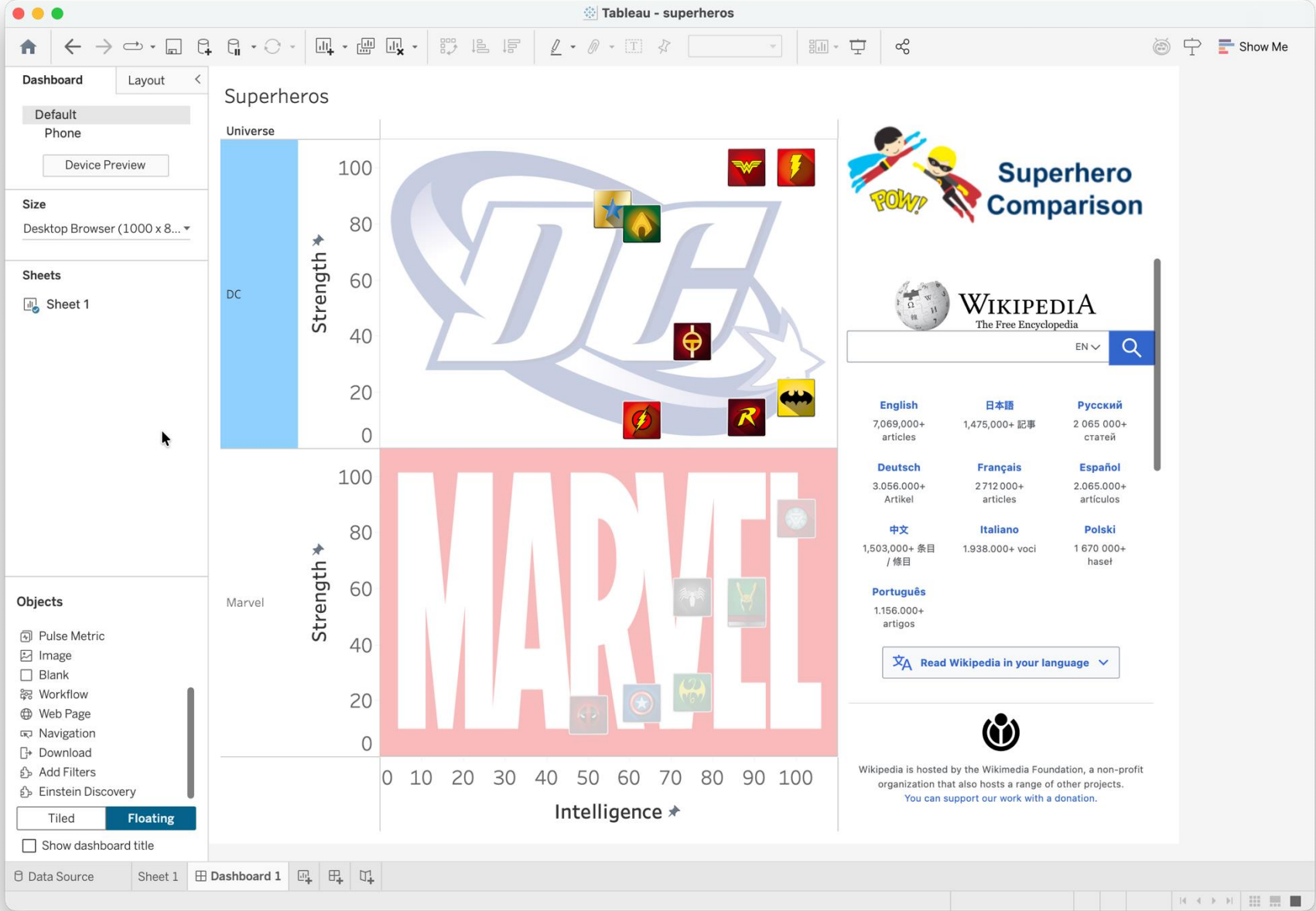
Cancel

OK

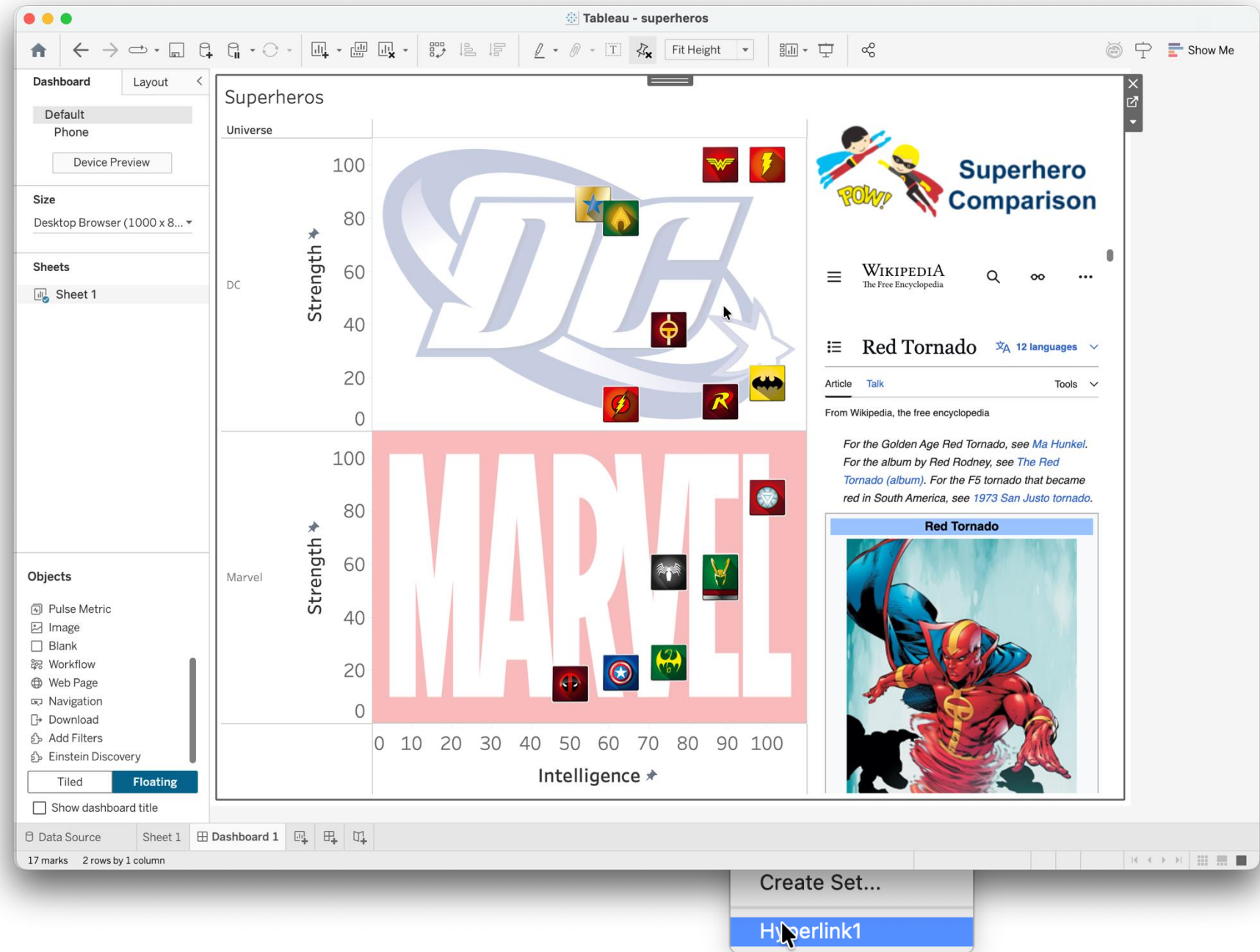
Load external data on click



Load external data on click



Load external data on click



More Resources

- **Need to modify your raw data?**
 - Ex. Add new variables, do more complex connections between datasets, address missing variables, etc.?
 - Try Tableau Prep: https://help.tableau.com/current/prep/en-us/prep_welcome.htm
- **Need additional resources for Tableau Desktop?**
 - Try the help pages (note the list in the left column of topics): <https://help.tableau.com/current/pro/desktop/en-us/default.htm>
 - Try data camp's free course: <https://www.datacamp.com/courses/introduction-to-tableau>
 - Try YouTube overviews: <https://www.youtube.com/watch?v=j8FSP8XuFyk>

Your Turn!

Find a group.

Download Students.zip from the course website. The README explains the datasets included. Choose at least 2 datasets, relate them in Tableau, and create a dashboard.

Not interested in Students data? Try any of these instead:

- NYC Flights:
<https://www.kaggle.com/datasets/aephidayatuloh/nyc-flights-2013>
- Fruit and Vegetable Prices: <https://www.ers.usda.gov/data-products/fruit-and-vegetable-prices>
- NPI Database:
https://download.cms.gov/nppes/NPI_Files.html